THE ARITHMETIC OF TIME

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Abstract. The fundamental laws are, to an excellent approximation, time-symmetric, yet the world runs one way. We present a small, self-contained mathematical model of time, formalized in Lean 4 (v4.35.0-rc2) against Mathlib in the public repository `github.com/stalex444/arithmetic-of-time`. The model is built from two polynomials: $x^3 - x - 1$, whose real root $\rho \approx 1.324718$ is the plastic number, and $x^4 - x - 1$, whose real root is $Q \approx 1.220744$. Five groups of kernel-verified theorems are proved. (1) Settling: the non-real roots z of $x^3 - x - 1$ contract at the exact rate $\rho\|z\|^2 = 1$, and on the ring $\mathbb{Z}[\rho]$ this sharpens to a Meyer-type separation inequality $|\sigma_1(x)| \cdot \|\sigma_2(x)\|^2 \geq 1$ for every nonzero x (`Time.meyer_separation`). (2) A boundary dichotomy: every non-real root of the cubic lies strictly inside the unit circle and every non-real root of the quartic strictly outside (`Time.pisot_boundary_dichotomy`), each by a single elementary Vieta reduction; thus ρ is a Pisot number and Q is not. (3) The mechanical word $m(n) = \lfloor (n+1)\beta \rfloor - \lfloor n\beta \rfloor$ of an irrational slope $\beta \in (0, 1)$ is two-valued, of density β, aperiodic, and balanced (`Time.balanced`). (4) An untunable chain: the constants $\kappa = (Q/\rho)^2$, $r_{\text{in}} = \sqrt{3}/2$, $\chi = Q/\rho$, and $A^* = \sqrt{2\rho} - \sqrt{3/4 - \rho^{-2}}$ satisfy the strict chain $\kappa < r_{\text{in}} < \chi < A^*$ (`Time.chain_strict`); the two thresholds are provably not algebraic integers while every monomial $\rho^a Q^b$ is an algebraic unit, so no such monomial can equal a threshold at any integer exponents. (5) Chirality: the Padovan substitution $a \mapsto b$, $b \mapsto c$, $c \mapsto ab$ has a factor language that is not closed under reversal: ca occurs, and ac occurs from no seed (`ArithmeticOfTime.padovan_chirality`); and every factor of length at least 4 of its seed-a language has its reversal absent, with 4 exact (`PDT.Padovan.exact_orientation_threshold_four`). On at most three letters the rule is forced: if a real eigenvalue $\theta > 1$ of a substitution's incidence matrix is a Pisot number, then $\rho \leq \theta$, with equality exactly for the Padovan substitution and its mirror $c \mapsto ba$, up to renaming the letters, and both are chiral (`ArithmeticOfTime.padovan_minimal_unique`, `ArithmeticOfTime.padovan_rules_chiral`). The substitution's matrix fixes both the scale ρ and the decay rate $\frac{1}{2}\log\rho$, and the direction costs exactly one bit more; a twelve-letter substitution realizing ρQ, at the least size any integer matrix with that eigenvalue allows and following Lind's construction, is chiral as well; and the Padovan history embeds as a future-directed timelike path in the $(3, 1)$ trace geometry of the quartic field, ordered alike by every future-directed observer. Every theorem depends only on the axioms `propext`, `Classical.choice`, and `Quot.sound`, with no `sorry` in any proof and no `native_decide`. A numerical companion computes the quasicrystalline diffraction of the mechanical word; the underlying pure-point theorem is cited, not formalized. We identify this arithmetic with time: the history is one completed object, and its direction is its handedness; the identification is stated at its tier, with each component at its own.

1. Introduction

The physics of time has a standard puzzle and a standard answer. The fundamental dynamical laws are, to an excellent approximation, the same whichever way time is run, yet the world runs one way. The usual answer puts the arrow not in the laws but in a boundary condition: the universe began in a state of extraordinarily low entropy, and everything one-way that we observe is the long relaxation away from it. Albert named this assumption the Past Hypothesis [2]; Penrose made vivid how special the required initial state must be [44]; Price and Carroll made the question of where the arrow comes from the organizing question of the subject [45, 17]. The hypothesis is itself contested [22], and there are published accounts that do without a special initial state [18, 10] or that find an arrow in geometry rather than in entropy [19]. This paper gives a different kind of answer. The model studied here is built from the plastic number $\rho \approx 1.3247$, the real root of $x^3 = x + 1$, and its laws are time-symmetric too: its single step, multiplication by ρ on $\mathbb{Z}[\rho]$, is invertible over the integers and conserves the field norm exactly (Section 3). But the history it produces, the word grown from one letter by the rule $a \mapsto b$, $b \mapsto c$, $c \mapsto ab$, is provably one-way: read backwards it is a different object, since ca occurs in it and ac never does, from any starting letter (Section 9). The input this costs is a single bit, the order of a and b in $c \mapsto ab$; the rule's matrix already fixes the rate of the accompanying contraction, $\frac{1}{2}\log\rho$ (Section 10); and every stretch of four letters already fixes which way the history runs (Section 9.2). What remains is still the selection of one object over its mirror image, and in that sense it has the shape of the Past Hypothesis. It is unlike it in content: the object selected is constructed, carries no parameters, and is asymmetric by theorem rather than improbable by measure.

The second point concerns relativity. In March 1955, writing to Michele Besso's son and sister shortly after Besso's death, Einstein wrote that the separation of past, present and future has only the meaning of an illusion, albeit a stubborn one [23].[1] The letter is often read as an endorsement of the block universe, in which past and future events are as real as present ones. That relativity *implies* such a picture is an argument, not a theorem: Rietdijk and Putnam argued for it [47, 46] and Stein answered them [55]; see [49]. What relativity does supply, invariantly, is a split. Which distant events count as simultaneous depends on the observer; the order of events along a timelike history does not [60], [29, Ch. 6]. Section 12 shows that the chiral history of this paper has exactly that split. The quartic field $\mathbb{Q}(Q)$, where $Q \approx 1.2207$ is the positive real root of $x^4 = x + 1$, carries a natural quadratic form, the trace form $\mathrm{Tr}(xy)$, and its signature is Lorentzian, $(3, 1)$. The history built from the cubic runs as a future-directed timelike path in that geometry, and the clock of *every* future-directed observer increases

[1] "…die Scheidung zwischen Vergangenheit, Gegenwart und Zukunft nur die Bedeutung einer wenn auch hartnäckigen Illusion." The English in the text is our translation.

along it. Observers may disagree about simultaneity; all of them read the history in the same order, with *ca* and never *ac*. That this arithmetic *is* time is the paper's identification, earned by coverage: the $(3, 1)$ trace form of the quartic is the signature of the history's geometry (Section 12), the tick word at slope β is its clock (Remark 6.11), and its direction is its handedness (Sections 9–10). The whole history already exists as one completed object, nothing in it flows, and what is experienced as forward motion is its handedness — the direction in which the history is the substitution's own language rather than its mirror's. The identification carries the evidential status of the ontology of any physical theory: each component keeps its own tier (Appendix A), it draws no laboratory prediction of its own, and its exposed surface is the one it inherits from its components.

The idea behind this paper came from chirality elsewhere in nature. The weak interaction is handed in its *law*: it acts only on left-handed fermions (and right-handed antifermions), a fact established within two years of the proposal that parity might fail [31] [58, 27, 28] and built into the $V-A$ theory [24, 56]. Living matter is handed in its *outcome*: proteins are built from L-amino acids and nucleic acids from D-sugars, although the underlying chemistry treats the two mirror forms alike [14].[2] If matter is handed at both of these levels, it is natural to ask whether time is handed in the same way. In the arithmetic studied here it is: time is handed in its *realized history*, while the laws stay symmetric. The analogy is closer than it may look. Parity and time reversal are different symmetries [11], but for a one-dimensional word, reflecting the line and reading the word backwards are the same operation, so at the level of the word, chirality *is* asymmetry under time reversal — reversal of the word, which is a different involution from inverting the tick (Section 10).[3]

Consider four structural demands one might place on a mathematical carrier of time.

(i) *Individuation*: states settle onto a discrete record, and distinct states are never confused.
(ii) *Irreversibility*: the settled history is one-way: read backwards, it is provably a different object.
(iii) *Aperiodicity*: the process never repeats.
(iv) *Non-randomness*: the succession of steps is maximally even rather than stochastic.

[2]The labels D and L name configurations, not directions of optical rotation. Parity violation separates the two mirror forms only by a very small energy difference, and whether the weak interaction seeds the biological preference is an open question [14, 48]; mechanisms that amplify a small initial excess are known [25, 52].

[3]The weak interaction does violate time-reversal symmetry (indirectly, through CP violation and CPT [20]; directly [7]), though, in Callender's words, these violations "don't appear to be responsible for thermodynamic behavior" [16]; the time-symmetry asserted here is that of the model's own laws.

This paper exhibits a concrete arithmetic structure meeting all four demands, with every claim machine-checked or explicitly labeled as cited, computed, or the identification. One convention is fixed here, once: the word *time* and its cognates in this paper (*moment*, *tick*, *arrow*) are names for the mathematical objects defined below; their identification with time is stated once, at the end of the second paragraph of this introduction.

The structure is built from two polynomials: $x^3 - x - 1$, whose real root $\rho \approx 1.324718$ is the plastic number [1] and, by a classical theorem of Siegel [51], the smallest Pisot number; and $x^4 - x - 1$, whose real root is $Q \approx 1.220744$. All results are formalized in Lean 4 (v4.35.0-rc2) against Mathlib in the public repository `github.com/stalex444/arithmetic-of-time`, or, where a registered entry or another public repository is named at the point of use, there; throughout, *kernel-verified* means proved there and checked by the Lean kernel, *classical* or *cited* means a literature theorem not formalized here, and *computed* means numerical evidence from the repository's `diffraction/` folder or an explicitly described finite computation. The results fall into five groups.

Settling and Meyer separation (demand (i)). For the non-real roots z of $x^3 - x - 1$, the transverse dynamics contracts at an *exact* rate: $\rho\|z\|^2 = 1$, so $\|z\| = \rho^{-1/2}$ — one exact step (`Time.tick_contraction`, `Time.norm_z_eq_rpow`). On the ring of integers $\mathbb{Z}[\rho]$ this sharpens to a Meyer-type separation inequality: writing σ_1 for the real and σ_2 for the complex embedding of $\mathbb{Q}(\rho)$, every nonzero $x \in \mathbb{Z}[\rho]$ satisfies $|\sigma_1(x)| \cdot \|\sigma_2(x)\|^2 \geq 1$ (`Time.meyer_separation`, `Time.meyer_separation_rpow`). Distinct arithmetic states can never be confused within a bounded window. The same theorems are re-proved by an independent second formalization (`TimeReplica.meyer_separation`, `TimeReplica.certification_soundness`).

The Pisot boundary: the cubic settles, the quartic spirals. Directly from the two defining polynomials, every non-real root z of $x^3 - x - 1$ satisfies $\|z\| < 1$ (`Time.cubic_conj_norm_lt_one`), while every non-real root w of $x^4 - x - 1$ satisfies $\|w\| > 1$ (`Time.quartic_conj_norm_gt_one`); together, $\|z\| < 1 \wedge 1 < \|w\|$ (`Time.pisot_boundary_dichotomy`). So ρ is a Pisot number and Q is not [12]. The proof is elementary and self-contained: writing $\sigma = w + \overline{w}$ and $s = w\overline{w} = \|w\|^2$ and reducing modulo the Vieta quadratic $w^2 = \sigma w - s$ yields one real identity per polynomial — $s^3 + s^2 = 1$ for the cubic, forcing $s < 1$, and $s^2 - \sigma^2 s = 1$ for the quartic, forcing $s > 1$ — one flipped inequality, driven purely by the degree. We are explicit about the tier boundary: this is the concrete settle/spiral dichotomy for these two polynomials; Siegel's theorem that ρ is the *smallest* Pisot number remains classical [51], as the required Pisot theory lies outside Mathlib; its restriction to Pisot numbers of degree at most 3 is kernel-verified (Section 8).

The tick: aperiodic but not random (demands (iii) and (iv)). For irrational $\beta \in (0, 1)$, the mechanical word $m(n) = \lfloor (n+1)\beta \rfloor - \lfloor n\beta \rfloor$ — a Sturmian object in the tradition of Morse and Hedlund [36, 34] — is kernel-verified to be *two-valued* (`Time.m_mem`), of *density* equal to the slope

(`Time.density_tendsto`, via the exact telescoping identity `Time.sum_m_eq`), *aperiodic* (`Time.aperiodic`), and *balanced*: the 1-counts of any two windows of equal length differ by at most one (`Time.balanced`). Balance is the precise "maximally even, not random" property of a Sturmian word. The irrationality of the slope, which lies beyond Mathlib, is carried as an explicit hypothesis throughout, never assumed as an axiom; for the slope β distinguished by this paper it is proved (`Time.beta_irrational`, Remark 6.10).

The untunable constant chain. The four algebraic constants $\kappa = (Q/\rho)^2$, $r_{\text{in}} = \sqrt{3}/2$, $\chi = Q/\rho$, and $A^* = \sqrt{2\rho} - \sqrt{3/4 - \rho^{-2}}$ satisfy the *strict* chain $\kappa < r_{\text{in}} < \chi < A^*$ (`Time.chain_strict`). Both threshold constants are proved *not* to be algebraic integers — r_{in} because $r_{\text{in}}^2 = 3/4 \notin \mathbb{Z}$ (`Time.rin_not_isIntegral`), and A^* by a cubic-field argument using no Galois machinery (`Time.Astar_not_isIntegral`) — while every monomial $\rho^a Q^b$ *is* an algebraic integer, indeed a unit (`Time.rho_norm_one`, `Time.Q_norm_neg_one`). Hence no monomial $\rho^a Q^b$ can ever equal a threshold, at any integer exponents (`Time.rin_ne_coupling_zmonomial`, `Time.Astar_ne_coupling_zmonomial`): the chain can never collapse into an algebraic coincidence.

The settled word is handed (demand (ii)). The Padovan substitution $a \mapsto b$, $b \mapsto c$, $c \mapsto ab$, whose incidence matrix has characteristic polynomial $x^3 - x - 1$, generates a factor language that reads differently backwards: ca occurs (in $S^{[5]}[a]$, hence from every seed, since $S^{[n]}[b] = S^{[n+1]}[a]$ and $S^{[n]}[c] = S^{[n+2]}[a]$) and ac occurs from no seed (`ArithmeticOfTime.padovan_chirality`), and no factor of length 4 or more is a palindrome (`ArithmeticOfTime.padovan_no_long_palindrome`). More is true: every factor of length at least 4 has its reversal absent from the language, and 4 is exact, because the palindrome aba occurs (`PDT.Padovan.exact_orientation_threshold_four`). We then ask four questions of this handedness. Is the rule itself forced? On at most three letters, yes, up to mirror image: the least Pisot eigenvalue an incidence matrix can have there is ρ, only s and its mirror $c \mapsto ba$ attain it, and both are chiral (Section 8). What does it cost? One bit: the rule's matrix fixes the scale and the decay rate, and only the order of a and b in $c \mapsto ab$ is left (Section 10). Does it survive at the scale ρQ, where the cubic and quartic arithmetic meet? Yes: a twelve-letter substitution realizing ρQ at its minimal size is chiral as well (Section 11). And is it compatible with relativity? The chiral history runs as a future-directed timelike path in the Lorentzian trace geometry of the quartic field, and every future-directed observer orders it the same way (Section 12).

These groups also have a *spectral* face, at a different tier. Laid out as positions and diffracted, the mechanical word shows sharp Bragg peaks supported on the incommensurate module $m + n\beta$ — quasicrystalline order — with a crystallographically forbidden peak growing proportionally to N while a Poisson point set of the same density stays flat. That companion

is *computed*, not proved; the underlying pure-point diffraction property is a theorem of the model-set literature [30, 50, 4] and is cited, never presented as formalized.

Scope. These are theorems about a mathematical *model* of time: individuation lives in the arithmetic of $\mathbb{Z}[\rho]$, the direction in the handedness of the settled word, the untunable constants in the geometry of the fields $\mathbb{Q}(\rho)$ and $\mathbb{Q}(Q)$, the tick in the symbolic dynamics of a mechanical word, and the settle/spiral dichotomy in the moduli of the roots. Beyond the identification stated above, we assert only the mathematics that the kernel has checked, at the tier at which it was checked. Every kernel-verified theorem's axiom dependencies form a subset of {`propext`, `Classical.choice`, `Quot.sound`} — the standard classical foundation of Mathlib — with no `sorry` in any proof (the registry's statement file `Challenge.lean` carries `sorry` placeholders by format), no `native_decide`, and no custom axiom anywhere in the development (audit in Section 14).

The paper proceeds as follows. Section 2 fixes notation and establishes the ground facts: irreducibility of the two polynomials and the norm computations exhibiting ρ and Q as units. Section 3 proves the exact contraction identity and the Meyer separation inequality. Section 4 proves the Pisot-boundary dichotomy. Section 5 proves the strict constant chain and the no-identity theorems that make it untunable. Section 6 develops the mechanical word: two-valuedness, density, aperiodicity, balance. Section 7 presents the computed diffraction companion. Section 8 proves that on at most three letters the Padovan substitution and its mirror are the only rules whose incidence matrix has the least Pisot eigenvalue, and that both are chiral. Section 9 proves that the Padovan language is chiral, locates the handedness of the tick word at its marked origin, and establishes the exact orientation threshold. Section 10 shows that the rate of the arrow is forced and its direction costs one bit. Section 11 constructs the twelve-letter substitution for ρQ and proves it chiral. Section 12 places the chiral history in Lorentzian signature. Section 13 records what is *not* proved — the frontier of the formalization. Section 14 describes the Lean development and the axiom audit. Appendix A gives the tier of every claim, Appendix B lists the declarations cited, and Appendix C records provenance.

2. Setup: two polynomials and their arithmetic

Fix once and for all the two polynomials

$$f_\rho(x) = x^3 - x - 1, \qquad f_Q(x) = x^4 - x - 1. \tag{1}$$

We write $\rho \approx 1.324718$ for the real root of f_ρ — the plastic number [1], the smallest Pisot number [51, 12] — and $Q \approx 1.220744$ for the unique positive real root of f_Q. Uniqueness in both cases is elementary and is recorded in Remark 2.4 below. Everything in this paper is arithmetic of these two algebraic units; from Section 3 onward, almost exclusively of the cubic one, until Sections 11–12, where the two are combined.

Theorem 2.1 (irreducibility)**.** *The polynomials $x^3 - x - 1$ and $x^4 - x - 1$ are irreducible in $\mathbb{Q}[x]$* (`Time.cubicQ_irreducible`, `Time.quarticQ_irreducible`)*.*

Both statements are kernel-verified. We sketch the formalized proofs; the route is reduction modulo 2.

Proof sketch. Both polynomials are monic over $\mathbb{Z}$, so it suffices to prove irreducibility of their images in $\mathbb{F}_2[x]$: irreducibility over $\mathbb{Z}$ follows, and then over $\mathbb{Q}$ by Gauss's lemma (both transfer steps are Mathlib lemmas about monic polynomials). Modulo 2 the polynomials become $x^3 + x + 1$ and x^4+x+1. A polynomial of degree 3 over a field is irreducible as soon as it has no root, and x^3+x+1 takes the value 1 at both elements of $\mathbb{F}_2$. The quartic also has no root in $\mathbb{F}_2$ (both values are 1), so a nontrivial factorization would have to be a product of two monic root-free quadratics; the only monic root-free quadratic over $\mathbb{F}_2$ is x^2+x+1, and $(x^2+x+1)^2 = x^4+x^2+1 \neq x^4+x+1$, the two sides differing in the coefficient of x. □

Consequently $K_\rho = \mathbb{Q}(\rho) \cong \mathbb{Q}[x]/(f_\rho)$ and $K_Q = \mathbb{Q}(Q) \cong \mathbb{Q}[x]/(f_Q)$ are number fields of degrees 3 and 4, and f_ρ, f_Q are the minimal polynomials of ρ, Q.

Theorem 2.2 (field norms)**.** *For $f \in \{f_\rho, f_Q\}$ let θ denote the class of x in $\mathbb{Q}[x]/(f)$, and let $N(\theta)$ denote the algebra norm over $\mathbb{Q}$, i.e. the determinant of the multiplication-by-θ endomorphism of $\mathbb{Q}[x]/(f)$ in the power basis $(1, \theta, \dots, \theta^{\deg f - 1})$. Then*

$$N(\theta) = 1 \quad \text{for } f = f_\rho, \qquad N(\theta) = -1 \quad \text{for } f = f_Q$$

(`Time.norm_ρ`, `Time.norm_Q`)*.*

Proof sketch. Kernel-verified. The formal proof exhibits the power basis of $\mathbb{Q}[x]/(f)$, of dimension 3 resp. 4, identifies the minimal polynomial of its generator with f (monicity plus Theorem 2.1), and applies the Mathlib identity "norm of the generator of a power basis $= (-1)^n$ times the constant coefficient of its minimal polynomial": here $(-1)^3(-1) = 1$ and $(-1)^4(-1) = -1$. □

Identifying $\mathbb{Q}[x]/(f_\rho)$ with K_ρ via $\theta \mapsto \rho$, Theorem 2.2 reads $N_{K_\rho/\mathbb{Q}}(\rho) = 1$ and $N_{K_Q/\mathbb{Q}}(Q) = -1$: the norms are units of $\mathbb{Z}$. The unit property itself is immediate from (1):

Proposition 2.3 (units)**.** *ρ is a unit of $\mathbb{Z}[\rho]$ and Q is a unit of $\mathbb{Z}[Q]$; explicitly*

$$\rho^{-1} = \rho^2 - 1, \qquad Q^{-1} = Q^3 - 1.$$

Proof. $\rho(\rho^2 - 1) = \rho^3 - \rho = 1$ and $Q(Q^3 - 1) = Q^4 - Q = 1$, directly from (1); this is consistent with the norms ± 1 of Theorem 2.2. □

Remark 2.4 (embeddings and signatures). (i) *Cubic.* Every real root r of f_ρ satisfies $1 < r$; this is kernel-verified (`Time.one_lt_root`), by the observation that $r \le 1$ makes $0 \le (1-r)(r+1)^2 = -r^2$, forcing $r = 0$, which is not a root. Since f_ρ is strictly increasing on $[1, \infty)$, the real root ρ is unique; and not all three complex roots can be real, since each real root exceeds 1 while the product of the roots is exactly 1 (elementary; not formalized). Hence K_ρ has signature $(1, 1)$: one real embedding and one conjugate pair of complex embeddings,

$$\sigma_1 : \rho \mapsto \rho \in \mathbb{R}, \qquad \sigma_2 : \rho \mapsto z, \quad \bar{\sigma}_2 : \rho \mapsto \bar{z},$$

where z denotes a non-real root of f_ρ. The choice within the conjugate pair is immaterial: every statement below is invariant under $z \mapsto \bar{z}$.

(ii) *Quartic.* Descartes' rule of signs applied to $f_Q(x)$ and to $f_Q(-x) = x^4 + x - 1$ gives exactly one positive and exactly one negative real root (elementary; not formalized): Q and a second real root Q', with $Q \in (1.22,\ 1.23)$ and $Q' \in (-0.73,\ -0.72)$ by exact rational sign evaluations of f_Q at the endpoints. The remaining two roots form a conjugate pair $w, \bar{w}$; the signature of K_Q is $(2, 1)$.

(iii) *Two meanings of "signature".* Above, signature means the number-field signature (r_1, r_2), the numbers of real embeddings and of conjugate pairs of complex ones. Sections 11–12 also use the signature of the trace form $(x, y) \mapsto \mathrm{Tr}(xy)$, a quadratic form over $\mathbb{Q}$: it is $(2, 1)$ for $\mathbb{Q}[x]/(x^3 - x - 1)$, $(3, 1)$ for $\mathbb{Q}[x]/(x^4 - x - 1)$, and $(7, 5)$ for their tensor compositum, all three kernel-verified in a registered entry [42] (`TraceForms.sigPos_cubic`, `TraceForms.sigNeg_cubic`, `TraceForms.sigPos_quartic`, `TraceForms.sigNeg_quartic`, `TraceForms.sigPos_compositum`, `TraceForms.sigNeg_compositum`).

Remark 2.5 (Pisot on the cubic side only). Theorem 3.1 below gives, kernel-verified, $\|z\| = \rho^{-1/2} < 1$: both non-real conjugates of ρ lie strictly inside the unit circle, so ρ is a Pisot number — the smallest one, by a classical theorem of Siegel [51] (cited, not formalized in general; its restriction to degree at most 3 is kernel-verified in Section 8; see also [12]). The quartic unit behaves differently. The product of the four roots of f_Q is its constant term, $Q \cdot Q' \cdot \|w\|^2 = -1$, so $\|w\|^2 = (Q\,|Q'|)^{-1} \ge (1.23 \cdot 0.73)^{-1} > 1$ by the brackets of Remark 2.4 (elementary; not formalized); numerically $\|w\| \approx 1.063$. Thus Q is *not* Pisot, and the separation mechanism of Section 3, which needs every conjugate other than the distinguished real one to contract, exists only on the cubic side.[4]

[4]For the quartic, the same single-window rule fails: the window contains points of $\mathbb{Z}[Q]$ whose transverse images are nonzero and arbitrarily close to 0 (`ChiralPadovanHistories.quartic_record_obstruction`, [39]). Windowing *both* real embeddings of $\mathbb{Q}(Q)$ restores a separation floor; that result is not part of this paper's formal perimeter.

We now fix the notation used throughout the rest of the paper. We work in the order $\mathbb{Z}[\rho] \subset K_\rho$ with power basis $(1, \rho, \rho^2)$, identifying $x = a+b\rho+c\rho^2$ with its coordinate triple $(a, b, c) \in \mathbb{Z}^3$, and we write

$$\sigma_1(a, b, c) = a + b\rho + c\rho^2 \in \mathbb{R}, \qquad \sigma_2(a, b, c) = a + bz + cz^2 \in \mathbb{C}.$$

Here and below $|\cdot|$ is the absolute value on $\mathbb{R}$ and $\|\cdot\|$ the modulus on $\mathbb{C}$. The formalization works with coordinate triples directly: for t in any commutative ring with $t^3 = t+1$ it sets `embed` t $(a, b, c) = a + bt + ct^2$ (`Time.embed`), and it does not presuppose that this representation is faithful — faithfulness is proved, from Theorem 2.1:

Lemma 2.6 (coordinate faithfulness)**.** *Let K be a field that is a $\mathbb{Q}$-algebra and let $t \in K$ satisfy $t^3 = t+1$. If $(a, b, c) \in \mathbb{Z}^3$ is nonzero, then $a+bt+ct^2 \neq 0$* (`Time.embed_ne_zero`).

Proof sketch. Kernel-verified. By Theorem 2.1 the minimal polynomial of such a t over $\mathbb{Q}$ is x^3-x-1, of degree 3; were $a+bt+ct^2 = 0$ with $(a, b, c) \neq 0$, the element t would be annihilated by the nonzero polynomial $a + bx + cx^2$ of degree at most 2. □

Remark 2.7 (hypothetical form of the statements). The formalized theorems of Section 3 do not construct ρ and z: they quantify over *every* pair $(r, z) \in \mathbb{R} \times \mathbb{C}$ with

$$r^3 = r + 1, \qquad z^3 = z + 1, \qquad \operatorname{Im} z \neq 0. \tag{2}$$

By Remark 2.4 such a pair exists and is unique up to $z \mapsto \bar{z}$, a replacement under which every statement below is invariant; the hypothetical form is therefore at least as strong as the same statement about the fixed pair (ρ, z). We state the theorems exactly as formalized.

3. The settling place: contraction and Meyer separation

In the model named in the introduction, one *tick* is multiplication by ρ on $\mathbb{Z}[\rho]$. On coordinates the tick is the integer map

$$(a, b, c) \longmapsto (c,\ a + c,\ b),$$

since $\rho \cdot (a + b\rho + c\rho^2) = c + (a + c)\rho + b\rho^2$ by (1) (`Time.tick`). Under either embedding the tick acts as multiplication by the corresponding root: `embed` t (`tick` p) $= t \cdot$ `embed` t p whenever $t^3 = t+1$ (`Time.embed_tick`), and by iteration $\sigma(\rho^n x) = \sigma(\rho)^n \sigma(x)$ for both embeddings (`Time.embed_tick_iter`). The line σ_1 carries expansion by $\rho > 1$ (Remark 2.4); the plane σ_2 — the *transverse plane*, the settling place of the title — carries multiplication by z. This section establishes the two metric facts about that plane: the contraction rate is an exact algebraic identity, and the transverse images of distinct points of $\mathbb{Z}[\rho]$ obey an explicit separation inequality.

Theorem 3.1 (exact contraction rate)**.** *Let $r \in \mathbb{R}$ and $z \in \mathbb{C}$ satisfy (2). Then*

$$r \cdot \|z\|^2 = 1$$

(`Time.tick_contraction`); *consequently* $\|z\| = r^{-1/2}$ (`Time.norm_z_eq_rpow`), *the exponent being the real power function on positive reals.*

Proof. This is the formalized proof, reproduced. Over $\mathbb{C}$, using $r^3 = r + 1$,

$$x^3 - x - 1 = (x - r)\left(x^2 + rx + (r^2 - 1)\right),$$

and $z \neq r$ because $\operatorname{Im} z \neq 0$; hence z is a root of the quadratic cofactor:

$$z^2 + rz + (r^2 - 1) = 0. \tag{3}$$

Conjugating (3) shows that $\bar{z}$ satisfies the same real quadratic, and $z \neq \bar{z}$; subtracting the two relations gives $(z - \bar{z})(z + \bar{z} + r) = 0$, hence $z + \bar{z} = -r$. Substituting back into (3) yields $z\bar{z} = r^2 - 1$. Therefore $\|z\|^2 = z\bar{z} = r^2 - 1$ and

$$r\|z\|^2 = r^3 - r = 1.$$

For the second form: $1 < r$ (`Time.one_lt_root`, Remark 2.4), so $r > 0$ and $\|z\| = (r^{-1})^{1/2} = r^{-1/2}$. □

Corollary 3.2 (strict contraction)**.** *Under* (2), $\|z\| < 1$ (`Time.norm_z_lt_one`).

Proof. $\|z\|^2 = r^{-1} < 1$ since $r > 1$. □

Theorem 3.1 is Theorem 2.2 made metric: the norm of ρ is its product over the three embeddings, $N_{K_\rho/\mathbb{Q}}(\rho) = \rho \cdot z \cdot \bar{z} = \rho\|z\|^2 = 1$. Each tick therefore scales the transverse plane by the *exact* factor $\rho^{-1/2}$ — not asymptotically and not up to a bounded error — and after n ticks by exactly $\rho^{-n/2}$, since $\sigma_2(\rho^n x) = z^n \sigma_2(x)$ (`Time.embed_tick_iter`). Because ρ is a unit (Proposition 2.3), the tick is moreover a bijection of $\mathbb{Z}[\rho]$ that conserves the field norm exactly (`PDT.NormFloor.N_tick`): the contraction is never blurred by iteration, and because its inverse is equally exact, the direction of time does not reside here (Section 9).

Theorem 3.3 (Meyer separation)**.** *Let $r \in \mathbb{R}$ and $z \in \mathbb{C}$ satisfy* (2)*, and let $p = (a, b, c) \in \mathbb{Z}^3$ be nonzero — equivalently, by Lemma 2.6, let $x = a + b\rho + c\rho^2$ be a nonzero element of $\mathbb{Z}[\rho]$. Write $x_1 = a + br + cr^2 \in \mathbb{R}$ and $x_2 = a + bz + cz^2 \in \mathbb{C}$. Then*

$$|x_1| \cdot \|x_2\|^2 \;\geq\; 1$$

(`Time.meyer_separation`); *equivalently, since $|x_1| > 0$,*

$$\|x_2\| \;\geq\; |x_1|^{-1/2}$$

(`Time.meyer_separation_rpow`).

Proof sketch. Kernel-verified; we sketch, indicating exactly what the formal proof supplies. The product of the images of p under the three embeddings is the norm form of K_ρ: with

$$N(a, b, c) = a^3 + b^3 + c^3 + 2a^2c + ac^2 - ab^2 - bc^2 - 3abc, \tag{4}$$

one has $x_1\, x_2\, \bar{x}_2 = N(a,b,c)$ (`Time.normForm_factor`). This is a polynomial identity in a, b, c, r, z modulo the two relations $r^3 = r + 1$ and (3); the formal proof certifies it by an explicit pair of cofactor multipliers against those relations (found by computer algebra, checked by the kernel), not by Galois-theoretic reasoning. Since $x_2\bar{x}_2 = \|x_2\|^2$, the integer $N(a,b,c)$ equals the real number $x_1\|x_2\|^2$. By Lemma 2.6 both $x_1 \neq 0$ and $x_2 \neq 0$, so $N(a,b,c)$ is a *nonzero* rational integer, whence

$$|x_1| \cdot \|x_2\|^2 = |N(a,b,c)| \geq 1.$$

The radius form follows by dividing by $|x_1| > 0$ and taking square roots. □

Remark 3.4 (independent second formalization). The results of this section are formalized twice. The file `SettlingReplica.lean` (namespace `TimeReplica`) re-proves Theorems 3.1 and 3.3 from scratch along an independently chosen route: the quadratic-cofactor step (3) is replaced by the Vieta relations of the full root triple (sum 0, pair relation, product 1), and the norm-form identity (4) is certified against the triangular system $\{\, u + v + t,\ v^2 + vt + t^2 - 1,\ t^3 - t - 1 \,\}$ (`TimeReplica.tick_contraction`, `TimeReplica.meyer_separation`, `TimeReplica.meyer_separation_rpow`). Both developments compile side by side against the same kernel, and nothing below depends on which copy is cited.

Remark 3.5 (what the two inequalities say). Apply Theorem 3.3 to a difference: if $x \neq y$ in $\mathbb{Z}[\rho]$ and $|\sigma_1(x) - \sigma_1(y)| \leq H$ for some $H > 0$, then

$$\|\sigma_2(x) - \sigma_2(y)\| \ \geq\ H^{-1/2}.$$

Distinct points of $\mathbb{Z}[\rho]$ whose σ_1-coordinates lie within a common window of width H keep transverse distance at least $H^{-1/2}$: within any bounded window, distinct integral states are never confused. This is the uniform-discreteness inequality underlying cut-and-project (model) sets built from Pisot units [30, 50, 4], here in completely explicit form with constant 1. The window hypothesis is essential and is carried explicitly through the formalization, because no unwindowed statement is true: the full image $\sigma_2(\mathbb{Z}[\rho])$ is dense in $\mathbb{C}$ (elementary; not formalized: its closure is a closed subgroup of $\mathbb{C}$ containing the nonzero elements $z^n = \sigma_2(\rho^n) \to 0$, so it is non-discrete; and if all its sufficiently small elements lay on one line through 0, then z^n and $z^{n+1} = z \cdot z^n$ would eventually both do so, forcing $z \in \mathbb{R}$). Theorem 3.1 complements the separation: the tick scales all transverse distances by exactly $\rho^{-1/2}$, so the contraction, once effected, is exact. The combined use of the two inequalities — a transverse reading certified within half the windowed separation retains the same nearest point of $\sigma_2(\mathbb{Z}[\rho])$ within the window after arbitrarily many ticks — is the subject of the next remark.

Remark 3.6 (certification soundness). Let (r, z) satisfy (2), let $H > 0$, let $w \in \mathbb{C}$ and $\ell \in \mathbb{Z}^3$ with $\|w - \sigma_2(\ell)\| < \frac{1}{2}H^{-1/2}$. Then for every n and every

$\ell' \neq \texttt{tick}^n(\ell)$ whose real image lies in the window, $|\sigma_1(\ell') - r^n\sigma_1(\ell)| \le H$,

$$\|z^n w - z^n\sigma_2(\ell)\| \;<\; \|z^n w - \sigma_2(\ell')\|$$

(`Time.certification_soundness`; independently `TimeReplica.certification_soundness`). Kernel-verified. In words: a reading certified within half the windowed separation of Remark 3.5 is carried by the tick to a reading whose nearest windowed lattice point is still the image of the original one, after any number of ticks. The proof combines Corollary 3.2 ($\|z^n\| \le 1$) with Theorem 3.3 applied to $\ell' - \texttt{tick}^n(\ell)$.

4. The Pisot boundary: the cubic settles, the quartic spirals

Each of the two defining polynomials has exactly one pair of non-real roots, and everything in this section is the answer to a single question: on which side of the unit circle does that pair lie? For $x^3 - x - 1$ it lies strictly inside; for $x^4 - x - 1$, strictly outside. Both statements are kernel-verified, by one and the same two-move argument, and the opposed inequalities are forced by the degree alone: the two trinomials have no adjustable coefficients, so nothing else could have decided the direction. The margins are modest — the moduli are $\rho^{-1/2} = 0.8688\ldots$ and $1.0633\ldots$ respectively, straddling 1 by barely thirteen and six percent — which is exactly why we prove the inequalities in the kernel rather than read them off a floating-point plot.

Definition 4.1 (Pisot and Salem numbers; classical)**.** A *Pisot number* is a real algebraic integer $\theta > 1$ all of whose Galois conjugates other than θ itself have modulus strictly less than 1. A *Salem number* is a real algebraic integer $\theta > 1$ whose conjugates other than θ all have modulus at most 1, at least one having modulus exactly 1 [12].

The kernel-verified statements are deliberately minimal. They quantify over an arbitrary complex number satisfying the defining equation with nonzero imaginary part; neither ρ nor Q appears in them, no polynomial is manipulated as an object, and no irreducibility, existence, or localization of roots is assumed. The formal proofs use only the field operations and complex conjugation on $\mathbb{C}$ — no root-finding and none of Mathlib's polynomial API.

Theorem 4.2 (the cubic settles)**.** *Let $z \in \mathbb{C}$ satisfy $z^3 = z + 1$ and $\operatorname{Im} z \neq 0$ — that is, let z be a non-real root of $x^3 - x - 1$. Then $\|z\| < 1$. Kernel-verified (`Time.cubic_conj_norm_lt_one`).*

Proof. Since $0^3 \neq 0 + 1$ we have $z \neq 0$. Set

$$\sigma = z + \overline{z} = 2\operatorname{Re} z, \qquad s = z\,\overline{z} = \|z\|^2.$$

Both are real, and $s > 0$ because $z \neq 0$. The pair $\{z, \overline{z}\}$ is the root set of the real quadratic $X^2 - \sigma X + s$, so

$$z^2 = \sigma z - s \tag{5}$$

(the Vieta quadratic). Reduce the defining equation modulo (5) by substituting twice:

$$z^3 = z \cdot z^2 = z(\sigma z - s) = \sigma z^2 - sz = \sigma(\sigma z - s) - sz = (\sigma^2 - s)\, z - \sigma s.$$

Subtracting $z^3 = z + 1$ gives

$$(\sigma^2 - s - 1)\, z + (-\sigma s - 1) = 0, \tag{6}$$

a linear relation on $\{1, z\}$ with *real* coefficients. Since $\operatorname{Im} z \neq 0$, the set $\{1, z\}$ is linearly independent over $\mathbb{R}$, so both coefficients vanish. (The formal proof executes this by conjugation: applying complex conjugation to (6) fixes the real coefficients and replaces z by $\overline{z}$; subtracting the two relations gives $(\sigma^2 - s - 1)(z - \overline{z}) = 0$, and $z \neq \overline{z}$.) Matching coefficients yields the real system

$$\sigma^2 = s + 1, \qquad \sigma s = -1. \tag{7}$$

Squaring the second identity and substituting the first,

$$1 = \sigma^2 s^2 = (s + 1)\, s^2, \qquad \text{i.e.} \qquad s^3 + s^2 = 1.$$

If $s \geq 1$ then $s^3 + s^2 \geq 2 > 1$, a contradiction; hence $\|z\|^2 = s < 1$ and $\|z\| < 1$. □

Theorem 4.3 (the quartic spirals)**.** *Let $w \in \mathbb{C}$ satisfy $w^4 = w + 1$ and $\operatorname{Im} w \neq 0$ — that is, let w be a non-real root of $x^4 - x - 1$. Then $1 < \|w\|$. Kernel-verified (`Time.quartic_conj_norm_gt_one`).*

Proof. As before $w \neq 0$; set $\sigma = w + \overline{w}$ and $s = w\,\overline{w} = \|w\|^2 > 0$, so that $w^2 = \sigma w - s$. Reduce w^4 modulo the Vieta quadratic, again by direct substitution:

$$\begin{aligned} w^4 = (w^2)^2 = (\sigma w - s)^2 &= \sigma^2 w^2 - 2\sigma s\, w + s^2 \\ &= \sigma^2(\sigma w - s) - 2\sigma s\, w + s^2 = (\sigma^3 - 2s\sigma)\, w + (s^2 - \sigma^2 s). \end{aligned}$$

Subtracting $w^4 = w + 1$ gives the real-coefficient relation

$$(\sigma^3 - 2s\sigma - 1)\, w + (s^2 - \sigma^2 s - 1) = 0,$$

and the $\mathbb{R}$-linear independence of $\{1, w\}$ — implemented by the same conjugate-and-subtract step — forces both coefficients to vanish:

$$\sigma^3 - 2s\sigma = 1, \qquad s^2 - \sigma^2 s = 1. \tag{8}$$

The first identity rules out $\sigma = 0$, since its left-hand side vanishes there; hence $\sigma^2 > 0$ and $\sigma^2 s > 0$. The second identity then reads $s^2 = 1 + \sigma^2 s > 1$, and since $s > 0$ we conclude $\|w\|^2 = s > 1$, i.e. $1 < \|w\|$. □

Theorem 4.4 (the Pisot boundary dichotomy)**.** *Let $z, w \in \mathbb{C}$ satisfy $z^3 = z + 1$, $w^4 = w + 1$, $\operatorname{Im} z \neq 0$, and $\operatorname{Im} w \neq 0$. Then*

$$\|z\| < 1 \quad \textit{and} \quad 1 < \|w\|.$$

Kernel-verified (`Time.pisot_boundary_dichotomy`).

Proof. The conjunction of Theorems 4.2 and 4.3; the formal declaration is exactly the pairing of the two proofs above. □

Remark 4.5 (one flipped inequality, driven by the degree). Set the two real systems side by side, each followed by what it forces on the single quantity $s = \|\cdot\|^2$:

$$\text{degree } 3: \quad \sigma^2 - s = 1, \quad \sigma s = -1 \qquad \Longrightarrow \quad s^3 + s^2 = 1 \qquad \Longrightarrow \quad s < 1;$$

$$\text{degree } 4: \quad \sigma^3 - 2s\sigma = 1, \quad s^2 - \sigma^2 s = 1 \qquad \Longrightarrow \quad s^2 = 1 + \sigma^2 s \qquad \Longrightarrow \quad s > 1.$$

The two proofs are the same two moves — reduce modulo the Vieta quadratic, match the real coefficients of the $\mathbb{R}$-linearly independent pair $\{1, w\}$ — applied to trinomials differing only in degree. The systems (7) and (8) are what degrees 3 and 4 respectively deposit, and they force opposite strict inequalities on the same quantity. The mechanism is visible in the constant coefficient of the reduction: at degree 4, squaring the Vieta quadratic squares its norm term, so the constant coefficient $s^2 - \sigma^2 s$ is quadratic in s with positive leading term, and equating it to 1 traps s above 1; at degree 3 the constant coefficient $-\sigma s$ is linear in s, and eliminating σ traps s below 1. No coefficient of either polynomial was available to tune; the direction of each inequality is a function of the degree alone.

Remark 4.6 (exact and numerical moduli). The systems pin the moduli exactly, although the theorems above do not need this. For the cubic, $t \mapsto t^3 + t^2$ is strictly increasing on $(0, \infty)$, so $s^3 + s^2 = 1$ has a unique positive solution; and $t = \rho^{-1}$ is one, since dividing $\rho^3 = \rho + 1$ by ρ^3 gives $\rho^{-3} + \rho^{-2} = 1$. Hence

$$\|z\| = \rho^{-1/2} = 0.8688\ldots$$

exactly, and $\sigma = -1/s = -\rho$, as the vanishing x^2-coefficient (the root sum) requires. For the quartic, eliminating σ from (8) shows that $s = \|w\|^2$ satisfies $(s^2 - 1)(s^2 + 1)^2 = s^3$, and numerically

$$\|w\| = 1.0633\ldots;$$

equivalently $\|w\|^2 = 1/(Q\,|r|)$, where $r = -0.7244\ldots$ is the second real root of $x^4 - x - 1$, because the product of the four roots equals -1. These identifications use knowledge of the real roots and are stated here in prose only; the kernel-verified statements are independent of them.

The passage from Theorem 4.4 to the Pisot conclusion requires three pieces of classical bookkeeping. They are elementary. Item (i) is Theorem 2.1, which is kernel-verified; items (ii)–(iii) we prove in prose and have *not* formalized — the kernel-verified statements above are deliberately independent of them.

Proposition 4.7 (classical bookkeeping; (ii)–(iii) not formalized)**.**

(i) *$x^3 - x - 1$ and $x^4 - x - 1$ are irreducible over $\mathbb{Q}$.*

(ii) *$x^3 - x - 1$ has exactly one real root, $\rho = 1.3247\ldots \in (1, 2)$; its remaining roots form a non-real conjugate pair $\{z, \overline{z}\}$.*

(iii) $x^4 - x - 1$ *has exactly two real roots,* $Q = 1.2207\ldots \in (1, 2)$ *and* $r = -0.7244\ldots \in (-1, 0)$*; its remaining roots form a non-real conjugate pair* $\{w, \overline{w}\}$.

Proof. (i) Both polynomials are monic with constant term -1, so a rational root would be ± 1, and neither is a root; this settles the cubic. For the quartic, a factorization over $\mathbb{Q}$ may be taken over $\mathbb{Z}$ by Gauss's lemma. If $x^4 - x - 1 = (x^2 + ax + b)(x^2 + cx + d)$ with $a, b, c, d \in \mathbb{Z}$, comparing x^3-coefficients gives $c = -a$, and then $bd = -1$, $a(d - b) = -1$, and $b + d = a^2$. From $bd = -1$ we get $\{b, d\} = \{1, -1\}$, so $b + d = 0$, forcing $a = 0$ and contradicting $a(d - b) = -1$.

(ii) The discriminant of $x^3 - x - 1$ is $-4(-1)^3 - 27(-1)^2 = -23 < 0$, so there is exactly one real root; it lies in $(1, 2)$ by the sign change $p(1) = -1 < 0 < 5 = p(2)$. The non-real roots of a real polynomial come in conjugate pairs, hence the remaining two form one such pair.

(iii) The derivative $4x^3 - 1$ of $q(x) = x^4 - x - 1$ has a single real zero, so q is strictly decreasing and then strictly increasing on $\mathbb{R}$ and has at most two real zeros; the sign changes $q(-1) = 1 > 0 > -1 = q(0)$ and $q(1) = -1 < 0 < 13 = q(2)$ produce exactly two, one in $(-1, 0)$ and one in $(1, 2)$. The remaining two roots form a non-real conjugate pair. □

Corollary 4.8. *ρ is a Pisot number. Q is neither a Pisot number nor a Salem number, and its failure of the Pisot condition is located entirely in the non-real pair: the second real conjugate satisfies $|r| < 1$.*

Proof. ρ is a real algebraic integer with $\rho > 1$; by Proposition 4.7(i)–(ii) its minimal polynomial is $x^3 - x - 1$, so its conjugates are z and $\overline{z}$, which satisfy $\|z\| = \|\overline{z}\| < 1$ by Theorem 4.2. This is Definition 4.1. Likewise Q is a real algebraic integer exceeding 1 whose conjugates are r, w, $\overline{w}$; Theorem 4.3 gives $\|w\| > 1$, so the Pisot condition fails, and since a conjugate lies strictly outside the closed unit disk the Salem condition fails as well. Finally $r \in (-1, 0)$ by Proposition 4.7(iii), so $|r| < 1$. □

Remark 4.9 (settle and spiral; classical, not formalized). The names in the section title record what the dichotomy does to powers. Let $t_n = \rho^n + z^n + \overline{z}^n$ be the n-th power sum of the roots of $x^3 - x - 1$: by Newton's identities, $t_{n+3} = t_{n+1} + t_n$ with $t_0 = 3$, $t_1 = 0$, $t_2 = 2$, so $t_n \in \mathbb{Z}$ for all n, and

$$\operatorname{dist}(\rho^n, \mathbb{Z}) \le |z^n + \overline{z}^n| \le 2\|z\|^n = 2\rho^{-n/2} :$$

the powers of ρ settle onto the integers at geometric speed. The power sums of the roots of $x^4 - x - 1$ are integers for the same reason, but the contribution of the non-real pair is $w^n + \overline{w}^n = 2\|w\|^n \cos(n \arg w)$ — a rotating term of expanding modulus. By the classical characterization of Pisot numbers, an algebraic integer $\theta > 1$ satisfies $\operatorname{dist}(\theta^n, \mathbb{Z}) \to 0$ if and only if θ is Pisot [12]; with Corollary 4.8 this gives $\operatorname{dist}(Q^n, \mathbb{Z}) \not\to 0$. The powers of Q spiral instead of settling. Everything in this remark is elementary prose or cited; none of it is formalized here.

One classical theorem remains deliberately outside the formal perimeter. Siegel [51] proved that ρ — the plastic number [1] — is the *smallest* Pisot number. We cite this and have not formalized it in general: minimality quantifies over the set of all Pisot numbers and is of a different character from the per-root inequalities proved above. Nothing downstream depends on it in general — later sections invoke only Theorem 4.4 and, in Section 8, its restriction to degree at most 3 — and its formalization is discussed with the rest of the frontier in Section 13. A neighbouring extremal property is now formalized: among monic irreducible integer polynomials of degree 3, resp. 4, with Mahler measure greater than 1, the polynomials $x^3 - x - 1$ and $x^4 - x - 1$ have the least Mahler measure (`MahlerWindow.cubic_min`, `MahlerWindow.quartic_min`; registered entry [40]). This is minimality of the measure in each degree, not uniqueness and not Siegel's theorem; and the measure of the quartic is not Q but $\psi \approx 1.380278$, the real root greater than 1 of $x^4 - x^3 - 1$ (`MahlerWindow.quartic_exact`), since the Mahler measure of $x^4 - x - 1$ is $Q\|w\|^2$. With the least measure in degree 2 from the same entry, which exceeds ρ, these minima do give Siegel's theorem in degree at most 3, because a Pisot number is the Mahler measure of its own minimal polynomial (`Time.Clock.mahler_eq_of_pisot`); that restriction is proved in Section 8.

5. An untunable chain of constants

The threshold analysis of the model turns on four constants built from the two roots. Two of them are monomials in ρ and Q; the other two are thresholds involving square roots. This section proves, first, that the four line up in a strict chain, and second, that the chain is rigid in a precise arithmetic sense: no monomial $\rho^a Q^b$ with integer exponents a, b can equal either threshold, so the strict inequalities can never degenerate into equalities under any retuning of exponents.

Definition 5.1. With ρ the real root of $x^3 - x - 1$ and Q the positive real root of $x^4 - x - 1$, set

$$\chi = \frac{Q}{\rho}, \qquad \kappa = \chi^2 = \left(\frac{Q}{\rho}\right)^2, \qquad r_{\text{in}} = \frac{\sqrt{3}}{2}, \qquad A^* = \sqrt{2\rho} - \sqrt{\tfrac{3}{4} - \rho^{-2}}.$$

Numerically $\kappa \approx 0.84919$, $r_{\text{in}} \approx 0.86603$, $\chi \approx 0.92151$,[5] and $A^* \approx 1.20326$; these approximations are quoted for orientation only — every proof below uses the exact defining equations and kernel-verified rational brackets. The constant r_{in} is the in-radius of the Voronoi cell of the unit 1 of $\mathbb{Z}[\rho]$ in the trace gauge fixed earlier, and A^* is the ambiguity threshold. The inner radicand of A^* is genuinely positive: $3/4 - \rho^{-2} > 0$, equivalently $3\rho^2 > 4$ (kernel-verified, `Time.radicand_pos`), so A^* is a difference of square roots of positive quantities and not an artifact of a convention for square roots of negative numbers.

[5] The quotient $\chi = Q/\rho$ is not the product ρQ of Section 11.

Theorem 5.2 (the strict chain)**.** $\kappa < r_{\text{in}} < \chi < A^*$ (`Time.chain_strict`).

Proof sketch. The kernel proof never compares square roots directly. Rational brackets $1.32 < \rho < 1.33$ and $1.22 < Q < 1.23$ are first established by sign evaluation of the defining polynomials at rational points (`Time.root_cubic_bounds`, `Time.root_quartic_bounds`). Each link is then squared against explicit positivity of both sides and reduced, via the defining equations, to a polynomial inequality that follows from the brackets: $\kappa < r_{\text{in}}$ is equivalent to $4(Q+1) < 3\rho^4$; $r_{\text{in}} < \chi$ is equivalent to $3\rho^2 < 4Q^2$; and $\chi < A^*$ follows from the rational cut $\chi < 0.94 < 1.18 < A^*$, the last inequality via $\sqrt{2\rho} > 1.62$ and $\sqrt{3/4 - \rho^{-2}} < 0.44$. □

Remark 5.3. The margins are not generous — $r_{\text{in}} - \kappa \approx 0.0168$ and $\chi - r_{\text{in}} \approx 0.0555$ — so strictness is a theorem, not an inspection. As throughout, the kernel statements quantify over every real r with $r^3 = r + 1$ and $r > 0$, and every real q with $q^4 = q + 1$ and $q > 0$; since each equation has exactly one positive real solution, these hypotheses pin $r = \rho$ and $q = Q$, and the quantified form is formally stronger than a statement about the fixed roots.

A strict inequality between fixed real numbers is, by itself, a fact about those numbers alone. The force of Theorem 5.2 lies in a second, structural statement. The two middle quantities are themselves members of the doubly indexed family $\rho^a Q^b$: indeed $\chi = \rho^{-1} Q$ and $\kappa = \rho^{-2} Q^2$. One may therefore ask whether the chain is an accident of these particular exponents — whether some other member of the family lands exactly on a threshold. It never does, at any integer exponents whatsoever, because the family and the thresholds are separated by an arithmetic dichotomy: every monomial is an algebraic integer, indeed a unit, while neither threshold is an algebraic integer at all.

Proposition 5.4 (monomials are unit algebraic integers)**.**

(1) *ρ and Q are algebraic integers, witnessed by the monic polynomials $X^3 - X - 1$ and $X^4 - X - 1$* (`Time.rho_isIntegral`, `Time.Q_isIntegral`).
(2) *$\rho^{-1} = \rho^2 - 1$ and $Q^{-1} = Q^3 - 1$, so both inverses are algebraic integers as well; consequently $\rho^a Q^b$ is an algebraic integer for all $a, b \in \mathbb{Z}$* (`Time.zpow_zmul_isIntegral`).
(3) *The field norms are $N_{\mathbb{Q}(\rho)/\mathbb{Q}}(\rho) = 1$ and $N_{\mathbb{Q}(Q)/\mathbb{Q}}(Q) = -1$, so ρ and Q are units in the rings of integers of their cubic and quartic fields* (`Time.rho_norm_one`, `Time.Q_norm_neg_one`, formalized for the canonical root of the defining polynomial in its abstract field $\mathbb{Q}[X]/(f)$).

Proof. (1) is the definition of an algebraic integer applied to the monic defining polynomials. For (2), the defining equations give $\rho(\rho^2 - 1) = \rho^3 - \rho = 1$ and $Q(Q^3 - 1) = Q^4 - Q = 1$, so each inverse is a polynomial in the root; the ring of algebraic integers is closed under sums, differences, products, and powers. For (3), the norm of a generator is $(-1)^{\deg f} f(0)$ for its monic minimal polynomial f, giving $(-1)^3 \cdot (-1) = 1$ for the cubic and $(-1)^4 \cdot (-1) = -1$

for the quartic. For general background on norms and units in the number rings attached to such roots see [12]. □

Proposition 5.5 (the lower threshold is not an algebraic integer)**.** $r_{\text{in}} = \sqrt{3}/2$ *is not an algebraic integer* (`Time.rin_not_isIntegral`).

Proof. If r_{in} were an algebraic integer, so would be its square $r_{\text{in}}^2 = 3/4$. But a rational number that is integral over $\mathbb{Z}$ lies in $\mathbb{Z}$, since $\mathbb{Z}$ is integrally closed in its fraction field, and $3/4 \notin \mathbb{Z}$. □

Theorem 5.6 (the upper threshold is not an algebraic integer)**.** $A^* = \sqrt{2\rho} - \sqrt{3/4 - \rho^{-2}}$ *is not an algebraic integer* (`Time.Astar_not_isIntegral`).

Proof. Three steps; the point of the route is that no Galois machinery is needed — only the integral closedness of $\mathbb{Z}$ and one degree count.

Step 1: $s = \sqrt{2\rho}$ *is an algebraic integer.* From $\rho^3 = \rho + 1$ we get $s^6 = (2\rho)^3 = 8\rho^3 = 8\rho + 8 = 4s^2 + 8$, so s is a root of the monic integer polynomial $X^6 - 4X^2 - 8$ (`Time.sqrt_two_mul_isIntegral`).

Step 2: reduction to the radicand. Suppose A^* were an algebraic integer. Then $\sqrt{3/4 - \rho^{-2}} = s - A^*$ would be a difference of algebraic integers, hence an algebraic integer, and so would be its square

$$v \;=\; \tfrac{3}{4} - \rho^{-2} \;=\; \rho^2 - \rho - \tfrac{1}{4},$$

where the second expression uses $\rho^{-2} = -\rho^2 + \rho + 1$, itself a consequence of $\rho^{-1} = \rho^2 - 1$ and the defining equation.

Step 3: v *is not an algebraic integer* (`Time.radicand_not_isIntegral`). Clearly $v \in \mathbb{Q}(\rho)$; conversely $\rho = (3 - 4v)/(4v + 1)$, the denominator $4v + 1 = 4\rho(\rho - 1)$ being nonzero, so $\mathbb{Q}(v) = \mathbb{Q}(\rho)$ and v has degree 3 over $\mathbb{Q}$. A direct computation from $\rho^3 = \rho + 1$ shows that v is a root of the monic cubic

$$X^3 - \tfrac{5}{4}X^2 + \tfrac{35}{16}X - \tfrac{23}{64},$$

which by the degree count is its minimal polynomial over $\mathbb{Q}$. If v were integral over $\mathbb{Z}$, then, $\mathbb{Z}$ being integrally closed, its minimal polynomial over $\mathbb{Q}$ would have integer coefficients; the coefficient $-5/4$ of X^2 is not an integer. This contradiction completes the proof. □

Theorem 5.7 (no monomial meets either threshold)**.** *For all* $a, b \in \mathbb{Z}$,

$$\rho^a Q^b \neq r_{\text{in}} \qquad \textit{and} \qquad \rho^a Q^b \neq A^*$$

(`Time.rin_ne_coupling_zmonomial`, `Time.Astar_ne_coupling_zmonomial`).

Proof. By Proposition 5.4 every $\rho^a Q^b$ is an algebraic integer; by Proposition 5.5 and Theorem 5.6 neither threshold is. An algebraic integer cannot equal a number that is not one. □

Remark 5.8 (the precise sense of "untunable"). Theorems 5.2 and 5.7 together say the following. The chain $\kappa < r_{\text{in}} < \chi < A^*$ is strict at the exponents at which the model presents it, and it cannot be made to collapse at any others: equality between a monomial $\rho^a Q^b$ and a threshold fails at

every pair $(a, b) \in \mathbb{Z}^2$. The separation is thus a dichotomy of arithmetic type — unit monomials on one side, non-integers on the other — rather than an inequality won by a numerical margin, and it is stable under every integer retuning of the exponents; the chain cannot collapse into an algebraic coincidence. Two sober qualifications are in order. First, the theorem forbids equality, not proximity: nothing here bounds how closely monomials may approach the thresholds. Second, the formalized statements concern the monomials $\rho^a Q^b$; the same dichotomy in fact excludes every element of the ring $\mathbb{Z}[\rho, Q]$, since that ring consists of algebraic integers, but this wider statement is classical and immediate rather than formalized in the accompanying repository.

6. The tick word: mechanical words are two-valued, dense, aperiodic, and balanced

Fix a real number β, the *slope*, and consider the integer sequence

$$m(n) \;=\; \lfloor (n+1)\beta \rfloor - \lfloor n\beta \rfloor, \qquad n \in \mathbb{N},$$

the *mechanical word* (Beatty word) of slope β [34, Ch. 2]. In the model naming fixed in the introduction, $m(n)$ is the duration of the n-th tick, and we call m the *tick word*. This section certifies, in the kernel, the four structural properties that make the tick word deterministic, structured, and non-repeating: it is *two-valued*; its running 1-count telescopes exactly, so its letter density is β; it is *balanced*; and, for irrational slope, it is *aperiodic*. All four are kernel-verified in the module `MechanicalWord.lean` (namespace `Time`) of the accompanying repository; a verbatim standalone restatement, free of the model naming, is provided in the namespace `Sturmian`. Every proof in the module reduces to the elementary floor bracket $\lfloor x \rfloor \le x < \lfloor x \rfloor + 1$ (in Mathlib: `Int.floor_le`, `Int.lt_floor_add_one`, and the characterizations `Int.le_floor`, `Int.floor_lt`); no symbolic-dynamics library is used.

We record precisely which hypotheses each statement carries. Two-valuedness assumes $0 < \beta < 1$; the telescoping identity, the density limit, and — notably — balance hold for *every* real slope, with no hypothesis at all; aperiodicity assumes β irrational, and irrationality enters the formal statement as an explicit hypothesis, carried rather than discharged (Remark 6.10).

Definition 6.1 (tick word and cumulative floor)**.** For $\beta \in \mathbb{R}$ let $m(n) = \lfloor (n+1)\beta \rfloor - \lfloor n\beta \rfloor$ for $n \in \mathbb{N}$ (`Time.m`) and let $C(N) = \lfloor N\beta \rfloor$ (`Time.cumFloor`). The word is the forward difference of the cumulative floor: $m(n) = C(n+1) - C(n)$ (`Time.m_eq_cumFloor`).

The engine behind both two-valuedness and balance is a one-line dichotomy for the floor of a sum.

Lemma 6.2 (floor-additivity dichotomy)**.** *For all $a, b \in \mathbb{R}$,*

$$\lfloor a + b \rfloor = \lfloor a \rfloor + \lfloor b \rfloor \quad \text{or} \quad \lfloor a + b \rfloor = \lfloor a \rfloor + \lfloor b \rfloor + 1.$$

(Time.floor_add_cases)

Proof. From $\lfloor a \rfloor \le a$ and $\lfloor b \rfloor \le b$ we get $\lfloor a \rfloor + \lfloor b \rfloor \le a+b$, hence $\lfloor a \rfloor + \lfloor b \rfloor \le \lfloor a+b \rfloor$. From $a < \lfloor a \rfloor + 1$ and $b < \lfloor b \rfloor + 1$ we get $a+b < \lfloor a \rfloor + \lfloor b \rfloor + 2$, hence $\lfloor a+b \rfloor \le \lfloor a \rfloor + \lfloor b \rfloor + 1$. An integer caught between s and $s+1$ equals s or $s+1$. □

Theorem 6.3 (two-valued)**.** *Let $0 < \beta < 1$. Then for every $n \in \mathbb{N}$, $m(n) = 0$ or $m(n) = 1$. (Time.m_mem)*

Proof. Since $(n+1)\beta = n\beta + \beta$, Lemma 6.2 with $a = n\beta$, $b = \beta$ gives $\lfloor (n+1)\beta \rfloor \in \{\lfloor n\beta \rfloor + \lfloor \beta \rfloor,\ \lfloor n\beta \rfloor + \lfloor \beta \rfloor + 1\}$, and $\lfloor \beta \rfloor = 0$ because $0 \le \beta < 1$. □

Thus for $0 < \beta < 1$ the tick word is a binary word: a two-letter alphabet, with no third symbol. Its 1-count is not merely asymptotically controlled — it telescopes *exactly*.

Proposition 6.4 (telescoping identity)**.** *For every $\beta \in \mathbb{R}$ and every $N \in \mathbb{N}$,*

$$\sum_{k<N} m(k) \;=\; \lfloor N\beta \rfloor.$$

(Time.sum_m_eq)

Proof. Induction on N: the empty sum is $0 = \lfloor 0 \rfloor$, and the inductive step is the forward-difference identity $m(N) = C(N+1) - C(N)$ of Definition 6.1. □

Theorem 6.5 (density)**.** *For every $\beta \in \mathbb{R}$,*

$$\frac{1}{N} \sum_{k<N} m(k) \;\longrightarrow\; \beta \qquad (N \to \infty).$$

(Time.density_tendsto)

Proof. By Proposition 6.4 the partial sum equals $\lfloor N\beta \rfloor$, and the floor bracket gives $\lfloor N\beta \rfloor \le N\beta < \lfloor N\beta \rfloor + 1$, recorded on the telescoped sum as `Time.sum_m_bounds`. Dividing by $N \ge 1$ yields $\beta - 1/N < \frac{1}{N}\sum_{k<N} m(k) \le \beta$, and the squeeze theorem concludes. □

For $0 < \beta < 1$, Theorem 6.5 says the frequency of the letter 1 in the tick word exists and equals the slope itself. Balance sharpens this from an asymptotic statement to a uniform one: *every* window of length L, wherever it sits, has 1-count within one integer of the ideal value $L\beta$. For a window of length L starting at position i write

$$S(i, L) \;=\; \sum_{k<L} m(i+k).$$

Lemma 6.6 (windowed telescoping and window membership)**.** *For every $\beta \in \mathbb{R}$ and all $i, L \in \mathbb{N}$:*

(1) $S(i, L) = C(i+L) - C(i)$ *(Time.window_sum);*

(2) $C(i+L)-C(i) = \lfloor L\beta \rfloor$ *or* $C(i+L)-C(i) = \lfloor L\beta \rfloor + 1$ *(`Time.window_mem`).*

In particular every window count $S(i, L)$ lies in the two-element set $\{\lfloor L\beta \rfloor,\ \lfloor L\beta \rfloor + 1\}$.

Proof. (1) is the telescoping of Proposition 6.4 shifted to start at i (induction on L). For (2), $(i+L)\beta = i\beta + L\beta$, so Lemma 6.2 with $a = i\beta$, $b = L\beta$ gives $\lfloor (i+L)\beta \rfloor - \lfloor i\beta \rfloor \in \{\lfloor L\beta \rfloor,\ \lfloor L\beta \rfloor + 1\}$. □

Theorem 6.7 (balanced)**.** *For every $\beta \in \mathbb{R}$ and all $L, i, j \in \mathbb{N}$,*

$$\Big| \sum_{k<L} m(i+k) \ - \ \sum_{k<L} m(j+k) \Big| \ \leq \ 1.$$

(`Time.balanced`)

Proof. By Lemma 6.6 both window counts lie in $\{\lfloor L\beta \rfloor,\ \lfloor L\beta \rfloor + 1\}$; two integers in a set $\{t, t+1\}$ differ by at most 1. □

Remark 6.8. Theorem 6.7 is the Sturmian *balance* property, and it is the strongest single statement of this section: any two equal-length factors of the tick word, taken anywhere in the word, have 1-counts differing by at most one, so no discrepancy accumulates at any scale. Note the total absence of hypotheses — the statement holds for every real slope, rational or not, inside or outside $(0, 1)$.

Balance says the word is maximally even; aperiodicity says it nevertheless never repeats.

Theorem 6.9 (aperiodic)**.** *Let β be irrational. Then the tick word has no period: there is no $p \in \mathbb{N}$ with $p > 0$ and $m(n+p) = m(n)$ for all $n \in \mathbb{N}$. (`Time.aperiodic`)*

Proof. Suppose $p \geq 1$ were a period. Since $m(n) = C(n+1) - C(n)$, periodicity of m says the forward differences of C at $n+p$ and at n agree for every n; induction on n then makes the cumulative floor additive over the period, $C(n+p) = C(n) + C(p)$, and a second induction gives linearity over multiples, $C(Mp) = M \cdot C(p)$ for all $M \in \mathbb{N}$. Write $c = C(p) = \lfloor p\beta \rfloor$ and let $\varphi = p\beta - c$ be the fractional part of $p\beta$. Then $\varphi \geq 0$, and $\varphi \neq 0$ because $p\beta$ is irrational (as $p \geq 1$ and β is irrational) and hence not an integer; so $\varphi > 0$. The upper floor bracket at Mp gives

$$Mp\beta \ < \ C(Mp) + 1 \ = \ Mc + 1, \qquad \text{i.e.} \qquad M\varphi < 1 \ \text{ for every } M \in \mathbb{N}.$$

By the Archimedean property there is an M with $M\varphi > 1$: a contradiction. □

Remark 6.10 (the irrationality hypothesis). In `Time.aperiodic` the irrationality of the slope is an explicit hypothesis of the kernel-verified statement — it is carried, never proved. All four theorems of this section quantify over an abstract slope β, so they attach to any particular slope the moment its

scope facts ($0 < \beta < 1$ for Theorem 6.3; irrationality for Theorem 6.9) are supplied from outside. For a concrete slope presented as a ratio of logarithms of multiplicatively independent algebraic numbers, irrationality is classical — it follows, for instance, from Baker's theorem on linear forms in logarithms — but no such result is available in Mathlib. For the slope distinguished by this paper (Remark 6.11) the version-2 repository discharges it without one: β is irrational (`Time.beta_irrational`), proved in the multiplicative form $\|w\|^{2n} \neq \rho^k$ for all k and all $n \geq 1$ (`Time.normSq_pow_ne_rho_pow`). The argument sends ρ to a non-real conjugate by a field embedding, which turns a putative equality into a product of three squared moduli of roots of $x^4 - x - 1$ equal to 1, and explicit bounds on the real roots exclude every such product; no Galois group is computed and no transcendence theory is used. Theorem 6.9 therefore holds at β. That $0 < \beta < 1$, the hypothesis of Theorem 6.3, is computed and not kernel-verified.

Remark 6.11 (the distinguished slope). The slope distinguished by this paper is

$$\beta \;=\; \frac{\log\|w\|}{\frac{1}{2}\log\rho} \;=\; \frac{\log(\psi/Q)}{\log\rho} \;=\; 0.4367860016\ldots,$$

where w is a non-real root of $x^4 - x - 1$ (Theorem 4.3) and ψ is the Mahler measure of $x^4 - x - 1$ (Section 4); the two forms agree because $\psi = Q\|w\|^2$. In words, β is the rate at which the quartic's non-real pair expands, measured in units of the rate at which the cubic's contracts (Theorem 3.1). The digits are computed. Taking the tick word at this slope as the model's clock is part of the identification stated in Section 1. The slope is a ratio of two magnitudes, and inverting the tick changes the sign of both logarithms; so β is unchanged by tick inversion and carries no direction. The direction of the model is located elsewhere (Section 9).

Remark 6.12 (Sturmian context; classical, not formalized). Theorems 6.7 and 6.9 are the forward half of the classical characterization of Sturmian words. By a theorem of Morse and Hedlund [36] (see also [34, Ch. 2]), an infinite binary word is Sturmian — equivalently, a mechanical word $n \mapsto \lfloor (n+1)\beta + x \rfloor - \lfloor n\beta + x \rfloor$ of irrational slope β and some intercept x; equivalently, a word of factor complexity $n + 1$ — if and only if it is aperiodic and balanced. Thus for irrational $0 < \beta < 1$, Theorems 6.3, 6.7 and 6.9 say precisely that the tick word is Sturmian; and conversely, by Morse–Hedlund, *every* aperiodic balanced binary word arises mechanically from an irrational slope. That converse is classical and is cited here, not formalized: the repository proves the four properties of m directly from the floor bracket and contains no part of the Morse–Hedlund equivalence. Formalizing the converse is listed among the open items in Section 13. Sturmian words are also one-dimensional model sets in the sense of aperiodic order [4], the point of view taken up when we turn to diffraction.

7. The spectral picture: quasicrystalline diffraction (computed)

Everything in this section is *computed*: it reports finite-N numerical evidence produced by the scripts in the `diffraction/` folder of the accompanying repository, and nothing in it is kernel-verified. The one general statement it rests on is a *cited* theorem of the literature, not a result of our formalization; we flag both facts again where they occur.

Let $\beta \in (0,1)$ be irrational and let $m(n) = \lfloor (n+1)\beta \rfloor - \lfloor n\beta \rfloor$ be the mechanical word of Section 6. Lay the word out as a point set on the line: the n-th gap is $1 + m(n) \in \{1,2\}$, rescaled so that the mean spacing is exactly $1/\beta$, and the positions $x_1 < x_2 < \cdots$ are the cumulative gap sums. We compare it against two point sets of *identical density* β: the periodic crystal $x_n = n/\beta$ and a Poisson point set of the same intensity (the "gas"). For a finite sample of N points the structure factor is

$$S(k) \;=\; \frac{1}{N}\left|\sum_{j=1}^{N} e^{ikx_j}\right|^2,$$

evaluated at $k = 2\pi\beta p$, where the reciprocal position p ranges over the module $\mathbb{Z} + \mathbb{Z}\beta$. For the crystal of period $1/\beta$ the diffraction is supported on integer p alone: $S(2\pi\beta p) = N$ for $p \in \mathbb{Z}$, and S vanishes at every incommensurate position $p = m + n\beta$ with $n \neq 0$. Those positions are *forbidden* to any periodic point set of that period.

The computation (slope $\beta = 0.4367860016$; any irrational slope exhibits the same phenomenon)[6] shows the mechanical word doing what no periodic set can: at $N = 200000$ it has sharp peaks throughout the incommensurate module — $S \approx 27543$ at $p = \beta$, $S \approx 175028$ at $p = 1 + \beta$, $S \approx 114558$ at $p = 2 + 2\beta$ — exactly where the crystal is identically zero, while the gas never exceeds 2.9 at any sampled position. (The peaks at $p = 1 + \beta$ and $p = 2 + 2\beta$ are those of the integer gap lattice, which every sequence of gaps 1 and 2 shares, and grow on the scale of N for that reason; the peak at $p = \beta$ is the word's own.) The hallmark of a genuine Bragg peak is intensity growing in proportion to N, whereas a diffuse spectrum stays $O(1)$. At the crystal-forbidden position $p = \beta$ the computed values are:

N	crystal	word	gas
25000	0.0	3059	1.18
100000	0.0	12478	0.06
400000	0.0	53114	0.33

The word's peak grows in proportion to N ($S/N \approx 0.12$–0.13), the gas stays $O(1)$, and the crystal has nothing there. Figure 1 shows the full comparison.

Remark 7.1 (Why this must happen — cited, not formalized). A mechanical word of irrational slope is a regular cut-and-project (model) set: the

[6] This is the slope β of Remark 6.11.

projection to the line of the points of $\mathbb{Z}^2$ lying in a strip of irrational slope. Regular model sets have pure-point diffraction. This is a theorem of Hof [30], extended by Schlottmann [50]; see Baake and Grimm [4] for a systematic treatment. It is classical: it is *not* formalized in the accompanying repository, and the computation above is finite-N evidence consistent with it, not a proof of it. The figure is the computation; the citation is the proof.

Thus the kernel-verified dichotomy of Section 6 — no period for any irrational slope (`Time.aperiodic`) yet window counts balanced to within one (`Time.balanced`) — is, spectrally, quasicrystalline order: aperiodic, but pure-point diffractive.

Remark 7.2 (Reproducibility). In the repository, `python3 diffraction.py` prints the peak table and the N-scaling above, and `python3 make_figure.py` re-renders Figure 1 (NumPy, plus Matplotlib for the figure). The run is deterministic (seed 0) except for the Poisson comparison set, whose flat spectrum is seed-independent.

8. The slowest settling clock on three letters

Siegel's theorem says that ρ is the smallest Pisot number. This section asks the same question of substitutions, and answers it in the kernel on alphabets of at most three letters: which rule grows a word at the slowest rate that still settles?

A *substitution* on an alphabet A sends each letter to a finite word over A; its *incidence matrix* has (i, j) entry the number of occurrences of the letter i in the image of j, and its *mirror* reverses every image. The Padovan substitution $s\colon a \mapsto b,\ b \mapsto c,\ c \mapsto ab$ of Section 9 has the incidence matrix M of Section 10, with characteristic polynomial $x^3 - x - 1$; its mirror $\tilde{s}$ differs only in $c \mapsto ba$. A real eigenvalue $\theta > 1$ of an incidence matrix is an inflation factor of the substitution (for the Perron eigenvalue, the factor by which iteration asymptotically lengthens words), and when θ is a Pisot number its remaining conjugates contract and its powers settle onto the integers (Section 4, Remark 4.9). We call such a θ a *settling rate* of the substitution, and a substitution attaining the least settling rate a *slowest settling clock*.

Theorem 8.1 (the slowest settling clock)**.** *Let σ be a substitution on $k \le 3$ letters, and let $\theta > 1$ be a real eigenvalue of its incidence matrix that is a Pisot number: every root of the minimal polynomial of θ other than θ lies strictly inside the unit circle. Then $\rho \le \theta$; and $\theta = \rho$ if and only if, after renaming the letters, σ is s or $\tilde{s}$* (`ArithmeticOfTime.padovan_minimal_unique`)*. The equivalence holds for every real eigenvalue θ of the incidence matrix, with no Pisot hypothesis* (`ArithmeticOfTime.padovan_eigenvalue_unique`)*. Neither primitivity nor the unit property is assumed.*

Proof sketch. Kernel-verified, in three steps. *The bound.* An eigenvalue of an integer $k \times k$ matrix is an algebraic integer of degree at most $k \le$

3. A Pisot number equals the Mahler measure of its minimal polynomial (`Time.Clock.mahler_eq_of_pisot`), and the least Mahler measures above 1 in degrees 2 and 3, from the registered entry [40], are at least ρ; in degree 1, θ is an integer at least 2. This is Siegel's theorem restricted to degree at most 3 (`ArithmeticOfTime.pisot_ge_plastic`). *The matrix.* If $\theta = \rho$, the characteristic polynomial of the incidence matrix, monic of degree $k \leq 3$ with integer coefficients, is divisible by $x^3 - x - 1$, so $k = 3$ and it equals $x^3 - x - 1$. A 3×3 matrix over $\mathbb{N}$ with that characteristic polynomial is M after one simultaneous permutation of its rows and columns (`ArithmeticOfTime.padovan_matrix_unique`): the trace 0 forces a zero diagonal, the coefficient of x leaves exactly one 2-cycle, and the determinant 1 leaves exactly one 3-cycle. *The substitution.* A matrix fixes each image as a multiset of letters. Two images of M are single letters; the third contains one a and one b, in one of two orders. The converse holds because s and $\tilde{s}$ both have the matrix M. □

Theorem 8.2 (both minimal rules are chiral)**.** *Under s, the word ca is a factor of $S^{[5]}[a]$, and ac is a factor of no iterate from any seed; under $\tilde{s}$, ac is a factor of the fifth iterate from a, and ca is a factor of no iterate from any seed* (`ArithmeticOfTime.padovan_rules_chiral`).

Proof. Kernel-verified. The statement for s is Theorem 9.2; the statement for $\tilde{s}$ follows from it because every iterate of $\tilde{s}$ from a one-letter seed is the reversal of the corresponding iterate of s (Proposition 10.1). □

Read together, the two theorems say this. On at most three letters the slowest settling clock is forced up to mirror image: its rate is ρ, and it is realized by exactly two rules, s and $\tilde{s}$. Both are chiral, so no slowest settling clock on at most three letters reads the same backwards; in that sense the arrow is forced. Which way it points is not: s and $\tilde{s}$ have the same matrix, the same rate and the same decay, and differ only in the order of the two letters in the image of c. That order is the one free bit, counted in Section 10. The open boundary is alphabets of more than three letters. There the bound $\rho \leq \theta$ still holds, by Siegel's theorem in general (cited), but uniqueness fails at once: adjoining a fourth letter $d \mapsto d$ to s keeps ρ as an eigenvalue. Whether every substitution whose incidence matrix carries only ρ's own spectrum (characteristic polynomial $x^m(x^3 - x - 1)$) is chiral is not decided here (Section 13).

The bound is not new: it is Siegel's theorem [51], which holds in every degree, and only its restriction to degree at most 3 is proved here. That the substitution with the fewest letters associated with the smallest Pisot number is unique up to reversal and relabelling was stated without proof by Mercat [35], whose substitutions carry no primitivity or non-erasing condition. What the kernel adds is a proof of the classification, over every substitution on at most three letters and every real eigenvalue of its matrix, and the chirality of both minimal rules.

9. The settled word is handed: chirality of the Padovan language

The tick word of Section 6 is binary. The settled structure of the cubic has a richer, three-letter description: the *Padovan substitution*

$$s\colon a \mapsto b, \qquad b \mapsto c, \qquad c \mapsto ab,$$

whose incidence matrix has characteristic polynomial $x^3 - x - 1$ (the fact is not used in the proofs of this section; it is load-bearing in Sections 8 and 10). Iterating s from a single letter grows the settled word of the cubic place. This section certifies, in the kernel, that the resulting factor language is *chiral*: it reads differently backwards, by theorem rather than by inspection of finitely many cases. The results of its first part are in `Chirality.lean` (namespace `Time.Padovan`); later parts name their modules where they occur.

Theorem 9.1 (the invariant)**.** *In every iterate $S^{[n+1]}[x]$ — any seed letter x, any n — every occurrence of a is immediately followed by b* (`Time.Padovan.every_a_followed_by_b`)*.*

The engine is one observation: a is produced only inside the block $s(c) = ab$, where b follows it within the block, and no block ends in a, so concatenation creates no violation (`good_append`, `good_S`, `good_iter`). Two consequences follow without enumeration.

Theorem 9.2 (chirality witness pair)**.** *The word ca is a factor of an iterate (it occurs in $S^{[5]}[a]$; `Time.Padovan.ca_factor`), while ac is a factor of no iterate from any seed* (`Time.Padovan.ac_not_factor`)*. Hence the factor language is not closed under reversal* (`Time.Padovan.chirality`)*.*

Theorem 9.3 (no long palindromes)**.** *No iterate $S^{[n]}[x]$, from any seed x and including $n = 0$, has a palindromic factor of length ≥ 4* (`Time.Padovan.no_long_palindromic_factor` for seed a; `ArithmeticOfTime.padovan_no_long_palindrome` for every seed)*.*

The proof of Theorem 9.3 runs through an explicit finite certificate: every factor of length ≤ 5 of every iterate lies in a 57-word list `F5` (`Time.Padovan.factors_complete`, via the window lemma `Time.Padovan.infix_window`: a factor of $S(w)$ of length L is contained in the image of a factor of w of length $\leq L$, because blocks are nonempty), and `F5` contains no palindrome of length 4 or 5 (`F5_no_palindrome`, by `decide`); a palindrome of length ≥ 4 contains a palindromic factor of length 4 or 5 (`palindrome_has_small_core`). The short palindromic factors that do occur can be read off the same certificate; they are exactly a, b, c, bb, cc, aba, bab, bcb, cbc. Completeness is proved for seed a; the invariant, the ac-avoidance and the exclusion of long palindromes cover every seed (the last because every iterate from seed b or c is a later iterate from seed a). We claim the kernel verification and the explicit certificate; the phenomenon type — substitution languages that fail reversal-closure —

is known, with published examples [6, Ex. 2] [8, Ex. 5.3], and palindrome *finiteness* follows in one line from non-reversal-closure for uniformly recurrent words by a published criterion [8, Thm. 3.2]. Reflection asymmetry for this substitution's tiling space was found independently, by a different method (asymptotic composants), in Gähler [26]; the prior study of this substitution's Rauzy fractal records that the substitution and its reversal have the same Rauzy fractal [35]. What is claimed as new here is the machine-checked proof and the certificate, not the phenomenon.

We emphasize the scope: these are statements about the factor *language*. Whether the associated subshift admits a *reversor* — a homeomorphism conjugating the shift to its inverse, the reversing-symmetry question of [6] — is a strictly stronger, conjugacy-level question, and language chirality alone does not decide it (there are reversible shifts with non-reversal-closed languages [6, Ex. 2]). No conjugacy-level claim is made in this repository.

9.1. **The marked origin of the tick word.** Returning to the binary tick word, extend it to integer indices: $sZ_\beta(n) = \lfloor (n+1)\beta \rfloor - \lfloor n\beta \rfloor$ for $n \in \mathbb{Z}$ (`Time.sZ` in `OriginDefect.lean`; `Time.sZ_natCast` ties it to `Time.m`). The word then satisfies an exact reflection identity about the origin,

$$sZ_\beta(-n-1) \;=\; sZ_\beta(n),$$

valid whenever $n\beta$ and $(n+1)\beta$ are not integers (`Time.sZ_reflect`; for irrational β this is every $n \notin \{0, -1\}$, `Time.sZ_reflect_irrational`; an unconditional form via ceilings is `Time.sZ_reflect_eq_ceil`). At the origin itself — the unique lattice hit $0 \cdot \beta \in \mathbb{Z}$ — the word carries a computable defect:

$$(sZ_\beta(-1),\, sZ_\beta(0)) = (1, 0) \qquad \text{for } 0 < \beta < 1$$

(`Time.origin_defect`). The reflection identity is the finitary, kernel-checked face of a structure known at the level of symbolic dynamics: for irrational slope, the two two-sided sequences at intercept zero form the distinguished asymptotic pair of the Sturmian system, each member palindromic away from the marked positions, and reversal of the marked orbit lands on the other member of the pair [9, Lemma 4.1]; that such pairs characterize the Sturmian sequences is Theorem A of Barbieri–Labbé–Starosta [9], proved there for all irrational slopes; hull-level reversibility of Sturmian shifts is Theorem 2 of [6]. Our statements are elementary and self-contained; the citations locate them in the published theory. The origin — and with it the handedness of the *marked* orbit, against the achirality of its hull — is where the model's asymmetry of time resides (see Section 1).

9.2. **The exact orientation threshold.** Theorem 9.3 says that a long factor is never its own reversal. A stronger statement is true: the reversal of a long factor is not anywhere in the language. Reading such a word backwards produces a word that the substitution never makes. Call a finite word u over $\{a, b, c\}$ *admissible* if it is a factor of $S^{[n]}[a]$ for some n.

Theorem 9.4 (exact orientation threshold)**.**

(a) *The palindrome aba is admissible:* $S^{[8]}[a] = cababbc$ (`PDT.Padovan.iterate_eight`, `PDT.Padovan.length_three_orientation_ambiguity`).

(b) *If u is admissible and has length at least 4, then the reversal of u is not admissible* (`PDT.Padovan.reversal_forbidden_of_four_le`).

Hence 4 is the least length from which every admissible word fixes the orientation (`PDT.Padovan.exact_orientation_threshold_four`).

Proof. Kernel-verified. Part (a) is a direct computation. For (b), by the certificate behind Theorem 9.3 every admissible word of length at most 5 lies in the list `F5`, and a finite decision shows that no word of length 4 in `F5` has its reversal in `F5` (`PDT.Padovan.F5_no_reversal_pair_length_four`, by `decide`). If u and its reversal were both admissible, then the first four letters t of u would be admissible, and so would the reversal of t, which is the last four letters of the reversal of u. Both would lie in `F5`, a contradiction. □

The two statements should be kept apart. Theorem 9.3 is the weak one: a long window is not symmetric. Theorem 9.4 is the strong one: a long window tells the direction, because its mirror image never occurs. Below length 4 there are exactly thirteen admissible words whose reversals are also admissible — a, b, c; ab, ba, bb, bc, cb, cc; aba, bab, bcb, cbc — nine palindromes and two mirror pairs (computed from the certificate). These are the only windows that cannot tell the direction. In the best case two letters suffice: the window ca fixes the direction, because ac is never admissible (Theorem 9.2). So every window of four letters decides, and some windows of two already do. The unit throughout is the *letter*: in the geometric history of Section 12 the three letters are steps of different lengths, so these are counts of letters, not durations.

A threshold of four is known in a different setting. For stationary binary sequences, the statistics of words shorter than four letters are automatically symmetric under reversal, so word-based measures of irreversibility cannot detect anything below length four [3]. Theorem 9.4 is a different kind of statement — exact rather than statistical, and about which words of a ternary language exist rather than how often — but in that vocabulary the Padovan language is maximally irreversible from length four on.

9.3. **Two handednesses, one arrow.** The paper now has two handed objects, and they live at different levels. The Padovan *language* is chiral: ac is absent while ca occurs (Theorem 9.2). The binary tick word is chiral in a subtler way: its language is closed under reversal, as the language of every Sturmian word is [34], so its hull is achiral; but its *marked* orbit, the one through the origin defect of Section 9.1, is not [9]. At the distinguished slope β of Remark 6.11 the irrationality this needs is kernel-verified (`Time.beta_irrational`; Remark 6.10); that $0 < \beta < 1$ is computed. The identification of Section 1 uses both: the arrow is the handedness of the marked orbit, and the past is read off the handedness theorems of the Padovan language — the

direction in which the settled word is the substitution's own language rather than its mirror's. Nothing in this paper identifies a letter of the Padovan word with a tick of the tick word, and we never call one a tick.

10. What the arrow costs: a forced rate and one bit

Sections 3 and 9 found two different things in the cubic arithmetic: a contraction at an exact rate, which is the same forward and backward, and a handedness, which is not. This section shows that both come from the same three-letter rule, and counts what the handedness costs.

10.1. **The rate is forced.** The incidence matrix of s, whose (i, j) entry counts the letter i in $s(j)$, is

$$M \;=\; \begin{pmatrix} 0 & 0 & 1 \\ 1 & 0 & 1 \\ 0 & 1 & 0 \end{pmatrix} \qquad \text{(rows and columns } a, b, c),$$

and it is exactly the matrix of the tick $(a, b, c) \mapsto (c,\ a + c,\ b)$ of Section 3. Its characteristic polynomial is $x^3 - x - 1$ and its determinant is 1. The determinant is the product of the eigenvalues, $\rho \cdot z \cdot \bar{z} = \rho\|z\|^2 = 1$, which is Theorem 3.1, kernel-verified for every root pair. So the decay rate of the non-Perron modes,

$$-\log\|z\| \;=\; \tfrac{1}{2}\log\rho \;=\; 0.1405997871\ldots,$$

is fixed by the same matrix that fixes the inflation factor ρ (for every real $\rho > 0$ and complex $\sigma \neq 0$ with $\rho\|\sigma\|^2 = 1$: `PDT.ArrowRate.rate_eq_half_log`, `PDT.ArrowRate.log_norm_eq_neg_half_log`). Scale and rate are the dominant and subdominant eigenvalues of one matrix, tied together by its determinant; neither is an independent input.

The rate is a magnitude, not a direction. Since $\rho^{-1} = \rho^2 - 1$, the inverse of the tick is again an integer matrix of determinant 1, and running the tick backwards exchanges expansion and contraction without changing the total: the topological entropy forward, $\log\rho$, equals the topological entropy backward, $2 \cdot \frac{1}{2}\log\rho$.

10.2. **The direction costs one bit.**

Proposition 10.1 (the two substitutions of the Padovan matrix)**.** *The substitutions on $\{a, b, c\}$ with incidence matrix M are exactly s and*

$$\tilde{s}\colon a \mapsto b, \qquad b \mapsto c, \qquad c \mapsto ba.$$

For every n and every seed x, the iterate $\tilde{S}^{[n]}[x]$ is the reversal of $S^{[n]}[x]$ (`PdtMirror.iterate_mir_seed`)*. Consequently the factor language of $\tilde{s}$ is the reversal of that of s: under $\tilde{s}$, ac occurs and ca never does.*

Proof. A matrix fixes each image as a multiset of letters. Here $s(a)$ and $s(b)$ are single letters, and $s(c)$ contains one a and one b, in one of two orders. For the second statement, $\tilde{s}$ is obtained from s by reversing every

image; that reversing every image reverses every iterate from a one-letter seed is a kernel-verified lemma about arbitrary substitutions on arbitrary alphabets. □

The two substitutions have the same matrix, hence the same inflation factor ρ and the same rate $\frac{1}{2}\log\rho$. They differ only in direction. Everything except the direction is paid for by the matrix; the direction costs exactly one bit, the order of the two letters in $s(c)$. That bit is a handedness. Any criterion that treats a word and its reversal alike gives s and $\tilde{s}$ the same verdict, so no mirror-symmetric principle can choose between them. Notice also that reversing the substitution leaves the matrix, and therefore the tick, unchanged, while reversing the tick replaces ρ by ρ^{-1}. Word reversal and tick inversion are different involutions, which is why the direction lives in the word and not in the tick.

The enumeration is not new. Mercat observed that, up to reversal and relabelling, there is a unique substitution with the fewest letters associated with the smallest Pisot number [35]; up to relabelling, s and $\tilde{s}$ are that substitution and its reversal. Theorem 8.1 proves the classification in the kernel for every substitution on at most three letters. What this section adds is the reading: the arrow costs one bit. One bit is very little beside an improbable initial state of the whole universe. It is not zero, and whether anything forces it is open (Section 13). So one three-letter rule supplies the scale ρ, the rate $\frac{1}{2}\log\rho$, and the handedness.

11. The substitution for ρQ: twelve letters, chiral

The cubic and the quartic meet in the product $\rho Q = 1.6171416\ldots$, which generates the degree-12 compositum $\mathbb{Q}(\rho, Q) = \mathbb{Q}(\rho Q)$ (registered: `PlasticThreshold.ruler_linearDisjoint`, `PlasticThreshold.ruler_conjugates_card`; [38]). Does the handedness of Section 9 survive at this scale? It does, with no input beyond the two founding substitutions.

11.1. **Construction and minimal size.** The quartic has its own substitution, $t\colon 1\mapsto 2,\ 2\mapsto 3,\ 3\mapsto 4,\ 4\mapsto 12$, whose incidence matrix has characteristic polynomial x^4-x-1 (direct computation). Writing p, q, r for a, b, c, the *row-major tensor* $s\otimes t$ acts on the twelve letters $\{p,q,r\}\times\{1,2,3,4\}$ by sending the pair (x,i) to the word $s(x)\times t(i)$ read in row-major order:

$$\begin{array}{llll} p1\mapsto q2, & p2\mapsto q3, & p3\mapsto q4, & p4\mapsto q1\,q2,\\ q1\mapsto r2, & q2\mapsto r3, & q3\mapsto r4, & q4\mapsto r1\,r2,\\ r1\mapsto p2\,q2, & r2\mapsto p3\,q3, & r3\mapsto p4\,q4, & r4\mapsto p1\,p2\,q1\,q2. \end{array}$$

We write $s_{\rho Q}$ for this substitution (`PdtRhoQ.s`).

Theorem 11.1 (characteristic polynomial)**.** *The incidence matrix of $s_{\rho Q}$ has characteristic polynomial*

$$P(x) \;=\; x^{12}-3x^9-2x^8+2x^6-x^5-3x^4-x^3+x-1$$

(`PdtCharpolyTensor.substitution_charpoly`; for the Kronecker product of the two founding matrices, `PdtCharpolyTensor.charpoly_tensor`). *The polynomial P is irreducible and is the minimal polynomial of ρQ* (registered: `PlasticThreshold.product_annihilator_iff`; `PDT.RulerMinpoly.ruler_minpoly`, `PDT.RulerMinpoly.rulerPoly_irreducible` in the same build).

The formal statement is about the substitution itself: the incidence matrix is computed by counting letters in the twelve image words, identified with the Kronecker product of the cubic and quartic matrices (`PdtCharpolyTensor.inc_eq_tensor`), and its characteristic polynomial is identified with P.

Corollary 11.2 (least size)**.** *Every square integer matrix having ρQ as an eigenvalue has size at least* 12*; the substitution $s_{\rho Q}$ attains this bound.*

Proof. The characteristic polynomial of such a matrix is a monic integer polynomial vanishing at ρQ, so it is divisible by the minimal polynomial P, of degree 12. □

Primitivity is computed, not kernel-verified (the tenth power of the incidence matrix is entrywise positive). Dominance is not merely computed: by Theorem 11.1 the eigenvalues are the conjugates of ρQ, and every conjugate other than ρQ has smaller modulus (registered: `PlasticThreshold.ruler_house`), so ρQ is the Perron eigenvalue.

The construction is Lind's: that products of Perron numbers are realized by products of shifts is [32, Thm. 2], [33, §6].[7] That a Perron number of degree d is realized by a nonnegative integer matrix of size d is not automatic: Perron numbers of negative trace need larger matrices [33, p. 289], and there are cubic Perron numbers whose least realizing size is arbitrarily large [59, Cor. 1.2]. Determining that least size in general is a named open problem: the minimum size problem of Boyle and Lind [15], stated for substitutions as Question 1 of Bertrand-Mathis and Nguema Ndong [13], whose Example 1 is the plastic number realized by the matrix M of Section 10. For ρQ the bound is attained, by Corollary 11.2; we claim nothing about the general question.

The quasicrystalline order of Section 7 belongs to the ρ side. The product ρQ is not a Pisot number: five of its conjugates, ρQ included, lie outside the unit circle (registered: `PlasticThreshold.ruler_conjugate_census`). By Solomyak's theorem [53], [54, Thm. 2.15], a one-dimensional self-similar tiling of a primitive substitution is weakly mixing exactly when its expansion factor is not Pisot, so the tiling system of $s_{\rho Q}$ (primitivity computed) is weakly mixing; since the pure-point part of its diffraction is carried by its

[7] For Perron numbers α, β one has $\mathbb{Q}(\alpha\beta) = \mathbb{Q}(\alpha)\mathbb{Q}(\beta)$ [33, Prop. 5]; coprimality of the degrees alone does not suffice, since $\sqrt[3]{2} \cdot e^{2\pi i/12}$ is a root of $x^6 + 4$, cf. [57]. The degree of ρQ is registered directly, by the irreducibility of P.

dynamical eigenvalues [21, 5], the diffraction has no nontrivial pure-point part.

11.2. **It is chiral.**

Theorem 11.3 (chirality of the ρQ substitution)**.** *Write $S_{\rho Q}$ for the letter-wise extension of $s_{\rho Q}$.*

(a) *The word $q2\,p1$ is a factor of $S_{\rho Q}^{[9]}[p3]$* (`PdtRhoQ.q2p1_factor`).
(b) *In every iterate $S_{\rho Q}^{[n+1]}[x]$, from every seed x, every $p1$ is immediately followed by $p2$* (`PdtRhoQ.every_p1_followed_by_p2`).
(c) *Hence for every letter $y \neq p2$, the word $p1\,y$ is a factor of no iterate from any seed* (`PdtRhoQ.p1_pair_not_factor`)*; in particular the factor language is not closed under reversal* (`PdtRhoQ.rhoQ_chirality`).

Proof. Kernel-verified, by the argument of Theorem 9.1, of which the formal proof is a port: $p1$ occurs in exactly one image, $s_{\rho Q}(r4) = p1\,p2\,q1\,q2$, at its start, and no image ends in $p1$, so every $p1$ in a concatenation of images is followed, inside the same image, by $p2$. □

Six of the forbidden words $p1\,y$ have reversals that do occur, namely those with $y \in \{q2, q3, q4, r2, r3, r4\}$. For $q2, q3, q4, r2$ the occurrence is kernel-verified (`PdtRhoQ.q2p1_factor`, `PdtRhoQ.q3p1_factor`, `PdtRhoQ.q4p1_factor`, `PdtRhoQ.r2p1_factor`); for $r3$ and $r4$ the shortest witnesses have 315 and 128 letters and are computed. The full picture is computed. On a stabilized prefix of 2,506,218 letters, the proportion of factors whose reversal is absent is 0, 41.77, 89.19 and 100 percent at lengths 1, 2, 3, 4; and 33 of the 79 two-letter factors are one-way. As for the Padovan word, every window of four letters decides the direction.

The handedness is not an artifact of the order chosen within the images. The incidence matrix fixes each image only as a multiset, and 768 substitutions realize it. All 768 are chiral (computed).

11.3. **Where the handedness comes from.** Each founding substitution has a chiral language: the cubic's by Theorem 9.2, the quartic's by computation (its two-letter factors 31 and 41 occur, while 13 and 14 do not). How the twelve-letter handedness divides between the two is open: the count depends on the order of letters within the images. Nor is inheritance automatic: the tensor of the Fibonacci substitution $a \mapsto ab$, $b \mapsto a$ with itself is chiral although the Fibonacci language is closed under reversal (computed). What does hold in general is the theorem below. And the orientation of the twelve-letter object is still a choice between mirror realizations, exactly as in Section 10: what costs nothing further is the *existence* of a handedness, not its direction.

11.4. **Mirror equivariance.** For a substitution σ, write $\overline{\sigma}$ for the substitution with every image reversed, and $\sigma \otimes \tau$ for the row-major tensor product.

Theorem 11.4 (mirror equivariance of the tensor)**.** *Let σ and τ be substitutions on arbitrary alphabets A and B. For every n and every seed $z \in A \times B$, the n-th iterate of $\overline{\sigma} \otimes \overline{\tau}$ from z is the reversal of the n-th iterate of $\sigma \otimes \tau$ from z* (`PdtMirror.iterate_tens_mir`)*. If every image of σ and of τ is a palindrome, then every iterate of $\sigma \otimes \tau$ is a palindrome and its factor set is closed under reversal* (`PdtMirror.palindromic_iterate`, `PdtMirror.palindromic_reversal_closed`)*.*

Kernel-verified, with no finiteness, decidability or primitivity hypothesis. In words: a tensor product cannot create handedness out of mirror-symmetric rules, and reversing both factors reverses the product. It holds for every pair of substitutions on every pair of alphabets and covers the row-major product only; we assert no novelty for it.

11.5. **One clock, two ticks.**

Remark 11.5 (the joint clock)*.* On the compositum $L = \mathbb{Q}(\rho) \otimes_{\mathbb{Q}} \mathbb{Q}(Q)$, multiplication by ρQ has no invariant rational subspace other than 0 and L (`PDT.JointClock.no_proper_jointTick_invariant_submodule`); yet the cubic and quartic ticks, multiplication by ρ and by Q, are explicit rational polynomials in it (`PDT.JointClock.joint_clock_decomposition`, `PDT.JointClock.recover_cubic_tick`, `PDT.JointClock.recover_quartic_tick`). By contrast, no nonzero additive map carries the cubic tick on $\mathbb{Z}^3$ into the quartic tick on $\mathbb{Z}^4$ (`PDT.CubicQuarticBridge.no_cubic_quartic_tick_intertwiner`). At the level of sequences, the termwise product of a sequence obeying $u_{n+3} = u_{n+1} + u_n$ with one obeying $v_{n+4} = v_{n+1} + v_n$ obeys the recurrence with characteristic polynomial P (registered: `PlasticThreshold.product_recurrence_charpoly`). So ρQ is one rationally indivisible clock that carries both ticks inside it.

12. A chiral history in Lorentzian signature

Everything in this section is kernel-verified in the registered entry PALOMAR-2026-09-04-000004 [39], which draws on the registered repository of this paper. We state the results and refer there for the proofs.

12.1. **The stream.** The iterates $S^{[n]}[a]$ are not nested, but every third one is: $U_n = S^{[3n]}[a]$ is a prefix of U_{n+1}, and together the U_n define one infinite word $\ell_0\ell_1\ell_2\cdots$ (`PDT.PadovanStream.U_eq_prefix`). Every finite block of it occurs in a genuine iterate; $\ell_6\ell_7 = ca$; and ac occurs nowhere (`ChiralPadovanHistories.admissible_stream`; `PDT.PadovanStream.block_factor`, `PDT.PadovanStream.letters_ca_at_six`, `PDT.PadovanStream.letters_no_ac`).

12.2. **The geometry.** Let $V = \mathbb{R}^4$ carry the quadratic form q with Gram matrix

$$G = \begin{pmatrix} 4 & 0 & 0 & 3 \\ 0 & 0 & 3 & 4 \\ 0 & 3 & 4 & 0 \\ 3 & 4 & 0 & 3 \end{pmatrix},$$

the trace form of $\mathbb{Q}[x]/(x^4-x-1)$ in the power basis (`ChiralPadovanHistories.trace_gram_link`), of signature $(3,1)$ (Remark 2.4). The vector $u_0 = (0,4,-3,0)$, that is $4Q - 3Q^2$, has $q(u_0) = -36 < 0$, and its orthogonal complement is a positive-definite three-dimensional rest space (`ChiralPadovanHistories.rest_space`). We call the timelike cone containing u_0 the *future*, and an observer *future-directed* if it lies in it. The map

$$B(t,w) = \big(\operatorname{Re} w/2,\ 4t,\ \operatorname{Im} w/2 - 3t,\ 0\big), \qquad t \in \mathbb{R},\ w \in \mathbb{C},$$

satisfies $q(B(t,w)) = \|w\|^2 - 36t^2$, and the clock of u_0 reads t on it.

12.3. **The history.** Fix ρ and a non-real root z of $x^3 - x - 1$. Give the three letters the steps

$$v_a = B(1,1), \qquad v_b = B(\rho, z), \qquad v_c = B(\rho^2, z^2),$$

and let H_n be the sum of the steps of the first n letters of the stream. The weights $(1, \rho, \rho^2)$ are not a choice: normalized letter weights compatible with the inflation, in the sense that each letter's weight times r is the total weight of its image, exist exactly when $r^3 = r + 1$, and are then $(1, r, r^2)$ (`ChiralPadovanHistories.normalized_inflation_iff`). This is the registered link from the substitution to ρ.

Theorem 12.1 (a chiral timelike history)**.** *There are $r > 1$ and a non-real z with $r^3 = r + 1$ and $z^3 = z + 1$ such that:*

(a) *every step $H_{n+1} - H_n$ is future timelike;*
(b) *H is injective and chronological: $H_n - H_m$ is future timelike whenever $m < n$;*
(c) *for every future-directed observer u, the clock of u strictly increases along H;*
(d) *the steps at positions 6 and 7 are those of c and a, and a step of a is never followed by a step of c;*
(e) *the first two steps are not collinear.*

(`ChiralPadovanHistories.exists_chiral_timelike_history`; the step code is injective and carries the chirality, `ChiralPadovanHistories.faithful_chiral_code`.) *Parts (a)–(c) hold for every $r > 1$ and every non-real root z of $x^3 - x - 1$* (`PDT.PadovanRealization.{future_increments, chronological, injective_history, every_future_observer}`).

So every future-directed observer reads the history in the same order. The direction is in the order, not in the endpoints: reversing a word leaves

the endpoint of its history unchanged and merely reverses the list of steps (`ChiralPadovanHistories.reversal_information`). Endpoint sums are abelian data, and abelian data cannot see the arrow. On the clock of u_0 the three steps have lengths 1, ρ and ρ^2; the letters are not equal units of time.

The arithmetic of Section 3 reappears along the history. Writing X_n and W_n for the real and complex coordinates of H_n — the sums of the weights $1, r, r^2$ and of $1, z, z^2$ over the first n letters — distinct positions satisfy

$$|X_n - X_m| \cdot \|W_n - W_m\|^2 \;\geq\; 1 \qquad (m \neq n)$$

for every sequence of letters (`ChiralPadovanHistories.history_separation`), the analogue of Theorem 3.3; and for $0 < \varepsilon \leq 1$, the history comes within ε of its starting transverse position only after clock time at least ε^{-2}, and does so by clock time $\rho^3\varepsilon^{-2}$ (`ChiralPadovanHistories.return_clock_bounds`).

12.4. **What the observers agree on.** The observer u_0 and the embedding B are choices; part (c) does not depend on the observer. That different observers disagree about which events are simultaneous is ordinary relativity [60], [29, Ch. 6]; what Theorem 12.1(c) adds is that, for this history, all of them agree on the order: the split described in the introduction.

13. Necessity: what is forced, what is cited, what is open

The kernel-verified results of this paper are sufficiency results. Sections 3, 6 and 9 together meet the four demands set out in the introduction; in particular, Section 6 proves that the word $m(n) = \lfloor (n+1)\beta \rfloor - \lfloor n\beta \rfloor$ meets demands (iii) and (iv): it is two-valued (`Time.m_mem`), its frequency of ones converges to β (`Time.density_tendsto`), it has no period when β is irrational (`Time.aperiodic`), and any two windows of equal length carry one-counts differing by at most one (`Time.balanced`). Section 4 proves that the two defining polynomials sit on opposite sides of the unit circle. A *necessity* statement — the four demands force this structure and no other — requires converses, and the converses are classical theorems that we cite but have not formalized. This section states the three classical characterizations that frame the necessity question, records exactly how far the kernel-verified material reaches into each, and estimates what a formalization of the remainder would cost. Nothing in the earlier sections depends on the unformalized converses; they mark the frontier.

13.1. **The Pisot boundary.**

Theorem 13.1 (Pisot; classical, cited, not formalized here)**.** *Let $\theta > 1$ be a real algebraic integer. Then $\operatorname{dist}(\theta^n, \mathbb{Z}) \to 0$ as $n \to \infty$ if and only if every Galois conjugate $\theta' \neq \theta$ of θ satisfies $|\theta'| < 1$. Such θ are the Pisot numbers* [12].

Section 4 kernel-verifies the concrete instance of the conjugate condition for the two polynomials of this paper: every non-real root z of $x^3 - x - 1$

satisfies $\|z\| < 1$ (`Time.cubic_conj_norm_lt_one`), every non-real root w of $x^4 - x - 1$ satisfies $\|w\| > 1$ (`Time.quartic_conj_norm_gt_one`), and the conjunction is packaged as a dichotomy (`Time.pisot_boundary_dichotomy`). Since $x^3 - x - 1$ has exactly one real root, the non-real pair exhausts the conjugates of ρ, so ρ meets Pisot's criterion; for Q, a single conjugate of modulus greater than one already disqualifies it. What is *not* formalized is Theorem 13.1 itself, in either direction: Mathlib, at the commit pinned in Section 14, contains no Pisot–Vijayaraghavan theory at all — not even the definition of a Pisot number — so the passage from the kernel-verified conjugate positions to the settling statement $\mathrm{dist}(\rho^n, \mathbb{Z}) \to 0$ is cited, not formalized. (The forward direction for the single number ρ is elementary — the power sum $\rho^n + z^n + \bar{z}^n$ is a rational integer satisfying $t_{n+3} = t_{n+1} + t_n$, and the conjugate terms decay geometrically — and would be a natural first extension of Section 4; it too is currently unformalized.)

13.2. **The Sturmian characterization.**

Theorem 13.2 (Morse–Hedlund; classical, cited, not formalized here)**.** *A sequence $s : \mathbb{N} \to \{0, 1\}$ is balanced and aperiodic if and only if there exist an irrational $\alpha \in (0, 1)$ and an intercept x such that s is the mechanical word $s(n) = \lfloor (n+1)\alpha + x \rfloor - \lfloor n\alpha + x \rfloor$ (or its upper variant) [36]; see also [34].*

The forward direction is kernel-verified in Section 6, in a form slightly more general than the classical statement: balance holds for *every* real slope (`Time.balanced` carries no irrationality hypothesis), and aperiodicity holds for every irrational slope, the irrationality entering as an explicit hypothesis as recorded there (`Time.aperiodic`). The converse is the necessity half — it asserts that the four demands are satisfiable *only* by mechanical words of irrational slope — and it is not formalized. A realistic estimate for formalizing it is 1200–2500 lines of Lean. The estimate is dominated by infrastructure: Mathlib today has no combinatorics on words whatsoever — no factor sets, no subword complexity function, no balance predicate — so every definition the converse quantifies over must be built first. On top of that, the proof needs three nontrivial lemmas:

(1) *Frequency existence.* A balanced sequence has a well-defined frequency of ones, obtained by a subadditivity argument; Fekete's lemma is available as Mathlib's `Analysis.Subadditive`, which shortens this step.
(2) *The rational case.* A balanced sequence whose frequency is rational is eventually periodic; hence aperiodicity forces the frequency to be irrational.
(3) *Intercept reconstruction.* Given a balanced aperiodic sequence of irrational frequency α, one must exhibit the intercept x realizing it as a mechanical word; the natural proof runs through equidistribution of the rotation by α on $\mathbb{R}/\mathbb{Z}$, where Mathlib's density theorems for `AddCircle` and its rotation-number theory give partial support.

The low end of the estimate covers the three lemmas over a minimal interface; the high end covers a reusable combinatorics-on-words development.

13.3. **Siegel minimality.**

Theorem 13.3 (Siegel; classical, cited, not formalized here)**.** *The smallest Pisot number is the real root* $\rho \approx 1.324718$ *of* $x^3 - x - 1$ [51].

This is the statement that upgrades "ρ qualifies" to "ρ is extremal among all qualifying bases." In general it is not formalized, and it is out of reach without a prior development of general Pisot theory: Siegel's argument is not a computation on one polynomial but an argument over the entire class of Pisot numbers, via rational functions with bounded integer coefficients, so nothing like the two-polynomial reduction of Section 4 applies. We do not attempt a line estimate. Its restriction to degree at most 3 is kernel-verified by a different route (`ArithmeticOfTime.pisot_ge_plastic`, Section 8): a Pisot number is the Mahler measure of its minimal polynomial, and the least Mahler measures are registered in degrees 2, 3 and 4. That route proceeds degree by degree and does not reach every degree.

Remark 13.4 (State of the necessity chain). Kernel-verified: the four demands are met (Sections 3, 6 and 9), and the conjugates of the two polynomials lie on the stated sides of the unit circle (Section 4). Cited: the converse of Theorem 13.2 (the demands force a mechanical word of irrational slope), Theorem 13.1 (settling is equivalent to the conjugate condition), and Theorem 13.3 (ρ is the least admissible base), whose restriction to degree at most 3 is kernel-verified (Section 8). Open, as formalizations: all three, Siegel's beyond degree 3. No claim in this paper is stated above these labels.

We close with the natural formalization targets, in increasing order of difficulty, offered as an invitation: each is a well-posed statement whose meaning is already fixed by the definitions of this paper.

(1) *Slope injectivity.* If $m_\beta = m_{\beta'}$ pointwise for $\beta, \beta' \in [0, 1]$, then $\beta = \beta'$: immediate from `Time.density_tendsto` by uniqueness of limits, a short exercise.
(2) *Frequency existence for balanced sequences.* Self-contained via `Analysis.Subadditive`; a few hundred lines.
(3) *The full Sturmian converse* (Theorem 13.2). The 1200–2500 line project above; it would seed a combinatorics-on-words library for Mathlib, with the complexity characterization of Sturmian sequences ($p(n) = n + 1$) as a natural milestone.
(4) *Pisot's theorem* (Theorem 13.1). Requires conjugate embeddings and algebraic-integer machinery, parts of which Mathlib's number-field library already provides; substantial but with a marked trail.
(5) *Siegel minimality in every degree* (Theorem 13.3). The summit: it presupposes the previous target together with a genuine theory of Pisot numbers. Degree at most 3 is done (Section 8).

The first two are within reach of a newcomer to Mathlib; the third would be a real contribution to the library; the last two would formalize a distinguished chapter of twentieth-century number theory. Any of them would be welcome.

Six further questions remain open.

(1) *Does anything force the bit?* Section 10 shows that no mirror-symmetric criterion can choose between s and $\tilde{s}$. Whether some other principle does is open.

(2) *Primitivity of $s_{\rho Q}$ in the kernel.* It is computed (Section 11); Perron dominance already follows from kernel-verified statements.

(3) *Alphabets of more than three letters.* Theorem 8.1 forces the rule, up to mirror image, on at most three letters. On larger alphabets the eigenvalue ρ is realized by many substitutions. Every one we tested whose incidence matrix carries only ρ's own spectrum (characteristic polynomial $x^m(x^3-x-1)$, equivalently every other eigenvalue inside the unit circle) is chiral: all 48 on four letters and all 480 on five. The achiral substitutions with Perron eigenvalue ρ that we found carry a second expanding factor: $a \mapsto b$, $b \mapsto c$, $c \mapsto d$, $d \mapsto e$, $e \mapsto f$, $f \mapsto g$, $g \mapsto bcacb$ has characteristic polynomial $(x^3 - x - 1)(x^4 + x^2 + x + 1)$ and palindromic images, so it realizes a different polynomial. Whether every substitution carrying ρ's own spectrum is chiral, on every alphabet, is open.

(4) *The split of the ρQ handedness.* How the one-way words of $s_{\rho Q}$ divide between the cubic and the quartic factor is open, and the computed split depends on the order of letters within the images.

(5) *A reversor for the Padovan shift.* Whether the Padovan subshift is conjugate to its inverse is a conjugacy-level question that language chirality does not decide (Section 9).

(6) *The thermodynamic arrow.* That the orientation arrow of this paper is the thermodynamic arrow is not claimed and is open.

The arrow of this paper is the handedness of Sections 9–12; what remains to be understood is how far that handedness reaches.

14. Formalization notes

The formal development accompanying this paper is available at `github.com/stalex444/arithmetic-of-time` (MIT license). It targets the toolchain `leanprover/lean4:v4.35.0-rc2` with Mathlib pinned at commit `065356127b1dc0016f66b7283ce0ce2c4055aa55` (recorded in `lean-toolchain` and `lake-manifest.json`), and `lake build` compiles the project modules in full: `Irreducibility`, `FieldNorm`, `Settling`, `SettlingReplica` (an independent second formalization of `Settling`), `ConstantChain`, `AlgebraicIntegers`, `MechanicalWord`, `PisotBoundary`, `Chirality` and `OriginDefect`; the registry files `Challenge` and `Solution`; and the modules added for version 2, listed in Appendix B, 44 modules in all.

Version 1 of the repository (commit `c48eb12`) is registered with the Palomar registry as PALOMAR-2026-08-19-000006 [37]. The registry compares three theorems, `ArithmeticOfTime.padovan_a_forces_b`, `ArithmeticOfTime.padovan_chirality` and `ArithmeticOfTime.padovan_no_long_palindrome`, over two compared definitions, `ArithmeticOfTime.s` and `ArithmeticOfTime.S`; the other modules are public in the repository but not compared. Version 2 of the repository consists of new commits to the same repository and is not a registry submission: the registered record stays at `c48eb12`, and the compared surface of version 2 (`comparator.json`, `Challenge.lean`, `Solution.lean`) is the same three theorems over `s` and `S`, unchanged. The uniqueness results of Section 8 are registered separately, as PALOMAR-2026-10-18-000019 [41], which compares `ArithmeticOfTime.padovan_minimal_unique`, `ArithmeticOfTime.padovan_eigenvalue_unique`, `ArithmeticOfTime.padovan_matrix_unique`, `ArithmeticOfTime.pisot_ge_plastic` and `ArithmeticOfTime.padovan_rules_chiral` over the definitions `ArithmeticOfTime.{s, mir, inc, sub}`; the same five statements, over the same definitions, are proved in this repository and not compared here. Every other result of version 2 is public in the repository and kernel-verified, but not compared. Appendix B lists all of them. Results cited from sibling entries are kernel-verified there: [39] (Section 12), [38] (Section 11), [42] (Remark 2.4) and [40] (Section 4); the results marked *public* in Appendix A are public in [43] but not registered.

The development satisfies the following audit, which is what the label "kernel-verified" means throughout this paper:

- no `sorry` in any proof; the registry's statement file `Challenge.lean` carries `sorry` placeholders by format, and each is discharged in `Solution.lean`;
- no `native_decide`: every decidable fact is checked by the Lean kernel itself, never delegated to trusted compiled code;
- no custom axioms: for every theorem, `#print axioms` reports a subset of {`propext`, `Classical.choice`, `Quot.sound`}, the standard classical foundation of Mathlib. The audit is reproducible declaration by declaration, e.g. `#print axioms Time.balanced`.

One design decision deserves emphasis: irrationality of the slope is never assumed. Every statement that requires β irrational carries `Irrational` β as an explicit hypothesis (as in `Time.aperiodic`); no axiom asserts the irrationality of any particular real number. The formal theorems are therefore exactly as conditional as their statements in this paper. For the distinguished slope the hypothesis is discharged by a theorem, `Time.beta_irrational`, not by an axiom.

The non-formal component of the repository is the `diffraction/` folder of Section 7, which is clearly marked as computation. The finite computations labelled *computed* in Sections 9 and 11 are described where they occur;

no Lean proof depends on any of them. Statements labelled "classical" or "cited" — for instance Siegel's theorem that ρ is the smallest Pisot number [51], and the pure-point diffraction of regular model sets [30, 50] — are literature theorems and are not formalized here.

Appendix A. Tiers

Every claim of the paper carries one of six tiers: **T0**, kernel-verified; **T1**, a cited theorem of the literature; **T2**, an explicit computation or elementary deduction not formalized; **T3**, sourced physics or philosophy; **T4**, the identification; **T5**, open. The "artifact" column says where a T0 result lives: *reg* = a compared statement of a registered entry; *closure* = in the build closure of a registered entry, not compared; *repo* = public in this paper's version-1 repository, and carried unchanged in version 2; *v2* = public in the version-2 repository, not compared; *public* = public, not registered (Appendix B lists the names).

claim	tier	artifact / source
Sections 2–6: every theorem so labelled	T0	repo (`c48eb12`)
Proposition 4.7(ii)–(iii)	T2	proved in prose
Siegel (in every degree), Pisot, Morse–Hedlund, model-set diffraction	T1	cited
Diffraction companion (Section 7)	T2	repo, `diffraction/`
Certification soundness (Remark 3.6)	T0	repo
Norm conservation by the tick (`PDT.NormFloor.N_tick`)	T0	v2
Rate forced, general form (`PDT.ArrowRate.*`)	T0	v2 (`rate_eq_half_log`); instance in repo
Least Mahler measure in degrees 3 and 4	T0	reg [40]
Distinguished slope β: digits; $0 < \beta < 1$	T2	computed
Irrationality of β (`Time.beta_irrational`)	T0	v2
Taking the tick word at slope β as the clock	T4	identification
Siegel's bound in degree at most 3 (`pisot_ge_plastic`)	T0	reg [41]; the theorem is [51]
Slowest settling clock on at most three letters: bound, uniqueness up to mirror image, matrix classification (Theorem 8.1)	T0	reg [41]
Both minimal rules are chiral (Theorem 8.2)	T0	reg [41]
Padovan invariant, chirality, no long palindromes, all seeds	T0	reg [37]
Inventory of short palindromes; the thirteen reversal-closed words	T2	read off the certificate
Exact orientation threshold 4 (Theorem 9.4)	T0	v2

claim	tier	artifact / source
Marked origin and its handedness (Section 9.1)	T0 + T1	repo; [9]; at slope β: irrationality T0 (v2), $0 < \beta < 1$ computed
Exactly two substitutions share M (Proposition 10.1)	T2	elementary; cf. [35]; up to renaming, T0 (Theorem 8.1)
Mirror substitutions have reversed languages	T0	v2 (`iterate_mir_seed`)
Characteristic polynomial of $s_{\rho Q}$	T0	v2
Minimal polynomial and irreducibility of ρQ; census; product recurrence	T0	reg [38]
Least size 12 (Corollary 11.2)	T2	standard algebra
Primitivity of $s_{\rho Q}$	T2	computed
Perron dominance of $s_{\rho Q}$ (Theorem 11.1 with `ruler_house`)	T0	v2 + reg [38]
Chirality of $s_{\rho Q}$ (Theorem 11.3)	T0	v2
Staircase 0/41.77/89.19/100; 33 of 79; 768 orderings	T2	computed
Quartic one-way pairs; Fibonacci tensor chiral	T2	computed
Split of the ρQ handedness	T5	open
Mirror equivariance of the tensor (Theorem 11.4)	T0	v2
Joint clock: irreducible, recovers both ticks; no intertwiner	T0	v2
Existence of the product construction; minimum size problem	T1	[32, 33, 15, 13, 59]
Trace-form signatures $(2,1)$, $(3,1)$, $(7,5)$	T0	reg [42]
Stream, geometry, chiral timelike history, observer agreement, separation	T0	reg / closure [39]
Multiplication by Q is not timelike-preserving	T0	public [43]
Raw ticks are not scalar Hamiltonian evolution	T0	public [43]
Non-Pisot side: the ρQ tiling is weakly mixing, no pure-point diffraction	T1 + T2	[53, 54, 21, 5]; primitivity computed
Simultaneity frame-dependent, timelike order invariant	T3	[60, 29]
Weak interaction, homochirality, T-violation facts	T3	cited in Section 1
Past Hypothesis and block-universe literature	T3	cited in Section 1
Time is this arithmetic (Section 1)	T4	identification
Thermodynamic arrow; a reversor; forcing of the bit; chirality on more than three letters for ρ's own spectrum	T5	open

Riders on the identification. The identification draws no laboratory prediction of its own; its clock and marked-orbit components are model-internal, and its exposed surface is the one it inherits from its components. The laws of the model are time-symmetric (the tick is a norm-preserving bijection with equal forward and backward topological entropy); the identification selects an object, it does not alter the laws. The residual input is a state selection — the shape of the Past Hypothesis — but the selected object is constructed, has no parameters, and is asymmetric by theorem. Word reversal and tick inversion are different involutions (Section 10). No claim of this paper rests on agreement between the slope β and other work in the author's programme.

Riders on Sections 11 and 12. The joint clock of Remark 11.5 is indivisible over $\mathbb{Q}$ only, not over $\mathbb{R}$ or $\mathbb{C}$, and the trace form of L keeps signature $(7, 5)$ (Remark 2.4). Theorem 12.1 is an existence and compatibility theorem, not a canonical selection: u_0, B, the normalization, the origin and the time orientation are explicit choices, and the image uses two spatial directions and time. Its geometry is the $(3, 1)$ trace geometry of $\mathbb{Q}(Q)$, not that of $\mathbb{Q}(\rho Q)$, whose trace form has signature $(7, 5)$. The substitution is not a Lorentz map: multiplication by Q sends u_0, with $q(u_0) = -36$, to Qu_0 with $q(Qu_0) = 91$ (`PDT.SignatureForcing.traceForm_tW`, `PDT.TimeStepTrace.traceForm_mul_tW`, `PDT.TimeStepTrace.timelike_to_spacelike`, `PDT.TimeStepTrace.not_preserves_timelike`; public in [43], not registered). Neither tick carried by the joint clock is a Hamiltonian evolution in the raw: a complex multiplier of modulus different from 1 is generated by no real scalar Hamiltonian, so neither the contracting cubic tick nor the expanding quartic tick is scalar Hermitian evolution (`PDT.HamiltonianBoundary.cubic_quartic_raw_ticks_not_scalarHamiltonian`; public in [43], not registered). That boundary is a statement about one complex mode; it does not turn the opposite radial directions of the two ticks into an arrow of time.

Appendix B. Declarations

Every declaration name in this paper exists in the Lean sources of the commit cited for it: names of this paper's repository in its version-2 sources, which carry every version-1 module unchanged in content; names of a sibling entry or of [43] at the commit given for it.

Compared in this paper's registered entry PALOMAR-2026-08-19-000006 [37] (`comparator.json`; the statements are in `Challenge.lean`, the proofs in `Solution.lean`; as registered at `c48eb12`, and unchanged in version 2). The compared definitions are `ArithmeticOfTime.{s, S}`: the Padovan substitution and its letterwise extension. The three theorems, in comparator order:

(1) `ArithmeticOfTime.padovan_a_forces_b`: in every positive iterate, from any seed, each a is followed by b (Theorem 9.1).

(2) `ArithmeticOfTime.padovan_chirality`: ca occurs and ac occurs from no seed (Theorem 9.2).
(3) `ArithmeticOfTime.padovan_no_long_palindrome`: no factor of length ≥ 4 of any iterate from any seed is a palindrome (Theorem 9.3).

Compared in the registered entry PALOMAR-2026-10-18-000019 [41]. The compared definitions are `ArithmeticOfTime.{s, mir, inc, sub}`: the Padovan substitution; the mirror, the incidence matrix and the letterwise application of a substitution on an arbitrary alphabet. The same five statements, over the same definitions, are proved in version 2 of this paper's repository (module `Supplement`) and are not compared there. The five theorems, in comparator order:

(1) `ArithmeticOfTime.padovan_minimal_unique`: on $k \leq 3$ letters a Pisot eigenvalue $\theta > 1$ satisfies $\rho \leq \theta$, with equality exactly for s and its mirror up to renaming (Theorem 8.1).
(2) `ArithmeticOfTime.padovan_eigenvalue_unique`: the same equivalence for every real eigenvalue, with no Pisot hypothesis (Theorem 8.1).
(3) `ArithmeticOfTime.padovan_matrix_unique`: a 3×3 matrix over $\mathbb{N}$ with characteristic polynomial x^3-x-1 is M up to one simultaneous permutation (Theorem 8.1).
(4) `ArithmeticOfTime.pisot_ge_plastic`: Siegel's bound $\rho \leq \theta$ for Pisot numbers of degree at most 3 (Section 8; not new [51]).
(5) `ArithmeticOfTime.padovan_rules_chiral`: s and its mirror are both chiral, in opposite directions (Theorem 8.2).

Proved in version 2 of this paper's repository, public, not compared (module `Supplement`, over the definitions of the two compared lists above, with the further definitions `ArithmeticOfTime.{tens, quart, rhoQSub, rhoQInc}`: the row-major tensor on arbitrary alphabets, the quartic substitution, the twelve-letter substitution and its incidence matrix).

- `ArithmeticOfTime.padovan_exact_orientation_threshold_four`: some factor of length 3 from seed a is its own reversal, and a factor of length ≥ 4 has its reversal in no iterate from any seed: the exact threshold 4 (Theorem 9.4).
- `ArithmeticOfTime.iterate_mir_seed`: every iterate of the mirror of any substitution, from a one-letter seed, is the reversal of the corresponding iterate (Proposition 10.1).
- `ArithmeticOfTime.iterate_tens_mir`: mirroring both factors of a row-major tensor reverses every iterate of the product (Theorem 11.4).
- `ArithmeticOfTime.single_mirrors_reverse`: the two single mirrors of a tensor give mutually reversed iterates (the companion of Theorem 11.4).

- `ArithmeticOfTime.padovan_length_three_orientation_ambiguity` and `ArithmeticOfTime.padovan_reversal_forbidden_of_four_le`: the two halves of the threshold theorem above (Theorem 9.4(a), (b)).
- `ArithmeticOfTime.rate_eq_half_log`: $\rho\|\sigma\|^2 = 1$ gives $-\log\|\sigma\| = \frac{1}{2}\log\rho$ (Section 10).
- `ArithmeticOfTime.rhoQ_substitution_charpoly`: the characteristic polynomial of the twelve-letter substitution is P (Theorem 11.1).
- `ArithmeticOfTime.rhoQ_chirality`: $q2\,p1$ occurs and $p1\,q2$ occurs from no seed (Theorem 11.3).

Compared in registered sibling entries. `ChiralPadovanHistories.{admissible_stream, faithful_chiral_code, exists_chiral_timelike_history, rest_space, trace_gram_link, normalized_inflation_iff, reversal_information, history_separation, return_clock_bounds, quartic_record_obstruction}` [39]; `PlasticThreshold.{ruler_linearDisjoint, ruler_conjugates_card, product_annihilator_iff, product_recurrence_charpoly, ruler_conjugate_census, ruler_house}` [38]; `TraceForms.{sigPos_cubic, sigNeg_cubic, sigPos_quartic, sigNeg_quartic, sigPos_compositum, sigNeg_compositum}` [42]; `MahlerWindow.{cubic_min, quartic_min, quartic_exact}` [40].

Public in sibling builds and in [43]*, not compared.* `PDT.RulerMinpoly.{ruler_minpoly, rulerPoly_irreducible}` in the build of [38] (and carried in the version-2 repository); `PDT.PadovanStream.{U_eq_prefix, block_factor, letters_ca_at_six, letters_no_ac}` and `PDT.PadovanRealization.{future_increments, chronological, injective_history, every_future_observer}` in the build of [39]; `PDT.SignatureForcing.traceForm_tW`, `PDT.TimeStepTrace.{traceForm_mul_tW, timelike_to_spacelike, not_preserves_timelike}` and `PDT.HamiltonianBoundary.cubic_quartic_raw_ticks_not_scalarHamiltonian` in [43].

Public in this paper's repository, not compared. All `Time.*` and `TimeReplica.*` names of Sections 2–9 other than those below, at `c48eb12` and in version 2. Carried in the version-2 repository (module: declarations):

- `SlowestClock`: `Time.Clock.{slowest_clock, clock_unique, matrix_unique, pisot_ge_rho, mahler_eq_of_pisot}`.
- `BetaIrrational`: `Time.{beta_irrational, normSq_pow_ne_rho_pow}`.
- `PdtFourTickThreshold`: `PDT.Padovan.{F5_no_reversal_pair_length_four, reversal_forbidden_of_four_le, iterate_eight, length_three_orientation_ambiguity, exact_orientation_threshold_four}`; with its dependency `PdtChirality` (the namespace-`PDT.Padovan` twin of `Chirality`).

- `PdtArrowRate`: `PDT.ArrowRate.{log_norm_eq_neg_half_log, rate_eq_half_log}`.
- `PdtNormFloor`: `PDT.NormFloor.N_tick`.
- `PdtMirrorTensor`: `PdtMirror.{iterate_mir_seed, iterate_tens_mir, palindromic_iterate, palindromic_reversal_closed}`.
- `PdtRhoQChirality`: `PdtRhoQ.{s, every_p1_followed_by_p2, p1_pair_not_factor, rhoQ_chirality, q2p1_factor, q3p1_factor, q4p1_factor, r2p1_factor}`.
- `PdtCharpolyTensor`: `PdtCharpolyTensor.{charpoly_tensor, substitution_charpoly, inc_eq_tensor}`.
- `PdtJointClock`, `PdtJointClockIrreducible`: `PDT.JointClock.{joint_clock_decomposition, recover_cubic_tick, recover_quartic_tick, no_proper_jointTick_invariant_submodule}`.
- `PdtCubicQuarticBridge`: `PDT.CubicQuarticBridge.no_cubic_quartic_tick_intertwiner`.
- `Sturmian`: the standalone restatement named in Section 6.

The repository also carries the supporting modules these import, among them the registered Mahler-measure modules of [40]; its `README.md` lists all 44 modules.

Appendix C. Provenance

Version 1 of this paper is dated 15 August 2026 and accompanies the repository at commit `c48eb12`, registered as PALOMAR-2026-08-19-000006. Version 2 adds Section 8, Sections 9.2–12, the new paragraphs of the introduction, and these appendices, and corrects the version-1 text where later results moved a fact; the principal changes are summarized in `paper/README.md` of the version-2 repository. The registered sibling entries cited are listed in the bibliography by identifier, version and commit. The reflection asymmetry of the Padovan tiling space is Gähler's [26]; the bound $\rho \leq \theta$ of Section 8 is Siegel's [51]; the minimal substitution for ρ is Mercat's [35]; the product construction is Lind's [32, 33]; the marked-orbit characterization is Barbieri–Labbé–Starosta's [9]; the reversing-symmetry framework is Baake–Roberts–Yassawi's [6].

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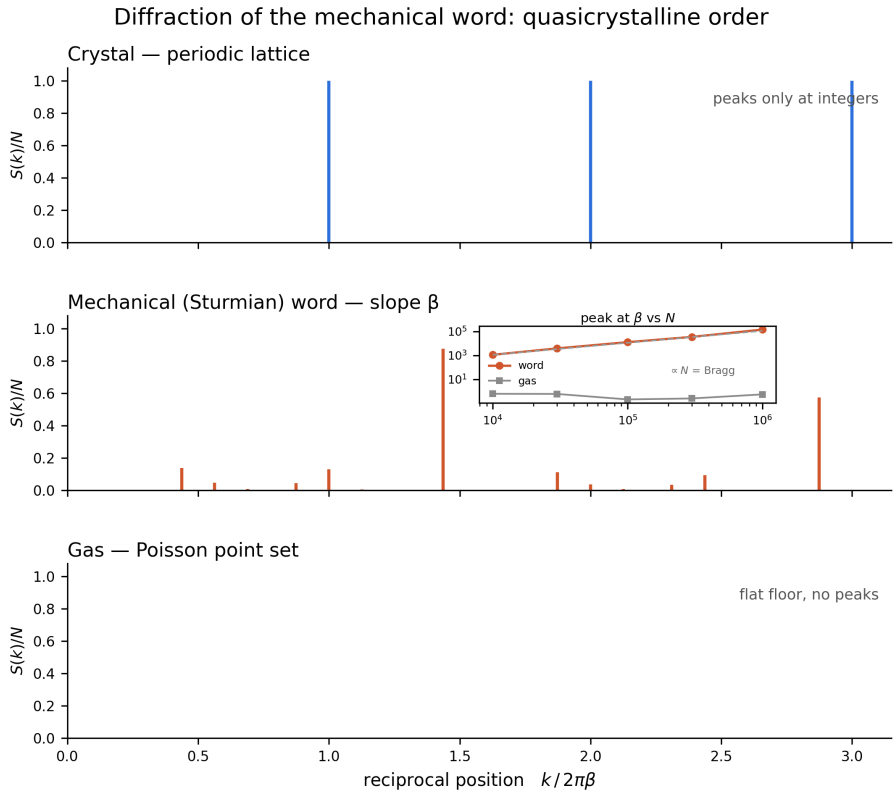
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Structure factor of three point sets at identical density $\beta = 0.4368$ (any irrational slope behaves the same). The mechanical word $m(n) = \lfloor (n+1)\beta \rfloor - \lfloor n\beta \rfloor$ has a pure-point spectrum — sharp Bragg peaks at the incommensurate module $m + n\beta$, forbidden to the crystal and absent in the gas. Inset: the incommensurate peak grows $\propto N$, a genuine Bragg peak. A mechanical word is a regular model set, so pure-point diffraction is a theorem (Hof 1995; Baake–Grimm 2013); this figure is the computation, not the proof.

FIGURE 1. Diffraction of the mechanical word (computed, not proved). Structure factor $S(k)/N$ for three point sets of identical density $\beta = 0.4367860016$, sampled at $k = 2\pi\beta p$ with p in the module $\mathbb{Z} + \mathbb{Z}\beta$, at $N = 200000$. Top: the periodic crystal $x_n = n/\beta$ — peaks at integer p only. Middle: the mechanical word $m(n) = \lfloor (n+1)\beta \rfloor - \lfloor n\beta \rfloor$ laid out as positions — sharp peaks at incommensurate positions $m + n\beta$, forbidden to the crystal and absent in the gas; the inset shows the intensity of the peak at $p = \beta$ growing in proportion to N (a genuine Bragg peak) while the gas stays $O(1)$. Bottom: the Poisson set — flat, no peaks. Every number here is a finite-N computation from the `diffraction/` folder; the pure-point statement it illustrates is a cited theorem [30, 50, 4], not a result of this paper's formalization.